



ARISTOTLE UNIVERSITY OF THESSALONIKI

FACULTY OF SCIENCE

PHYSICS DEPARTMENT

PROGRAM OF UNDERGRADUATE STUDIES

ELECTRONICS AND COMPUTERS

AN ANALOG INTEGRATED CIRCUIT OF A CHARGE
SENSITIVE PREAMPLIFIER FOR HIGH ENERGY
PHYSICS EXPERIMENTS

BACHELOR THESIS

by

ANASTASIOS KAZAKOS

Supervising Professor: Thomas Noulis

February 22, 2024

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Acknowledgements

I consider the decision of taking my optional thesis as one of the best in my undergraduate journey. Although the process was unprecedented and the difficulties encountered were unusual, the gain in knowledge and familiarity with research and reporting was of great significance. In completion of this journey, I realized the value of the guidance and support of certain individuals, for which I would like to express my gratitude.

Firstly, I would like to warmly thank my thesis supervisor, Mr. Thomas Noulis. His constructive criticism on the topics that make up this thesis greatly determined its final form. Additionally, I would like to thank Vasiliki Gogolou, a postdoctoral researcher at AUTH, for her significant contribution to my familiarity with the technical part of the thesis. Her support greatly facilitated the completion of the technical part, saving me considerable time. I also express my thanks to all the professors of the undergraduate courses in the field of electronics and computers in the Department of Physics, who played an important role in laying my foundations in this field.

Finally, I would like to thank my family for all the support they have given me during my studies. Their presence was valuable in accomplishing my goals.

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Abstract

The subject of this thesis is the study and the implementation of a Charge Sensitive Preamplifier (CSA) for capacitive detection in High Energy Physics experiments [HEP]. The design was implemented using a 0.18 μm CMOS process by XFAB. To simulate the capacitive detector, a current source in parallel with a capacitor was used. It is worth mentioning that the CSA circuit provides an output conversion gain of 2.29 mV/fC, a total gain of 122 dB, and a bandwidth of 505 kHz.

Five major chapters are the basis for this bachelor thesis. In the first chapter, the applications of CSA and different groups of detectors are briefly discussed. In addition, an introduction takes place about the basic characteristics of the CSA as well as the description of its operation. Emphasis is given to Microstrip Detectors and Pixel Detectors as they are the most common types in HEP experiments. In particular, not only the materials that are used for semiconductor detectors are discussed, but charge generation and charge transport inside the detector as well.

The second chapter concerns the noise. At first, the mathematical tools for noise analysis are described. Later, the most common types of noise are explained and noise sources on CMOS technology are presented. Emphasis is given to thermal and flicker noise, as they contribute mostly to this study.

In the third chapter, an analysis takes place for the amplification topology of the Charge Sensitive Preamplifier. Initially, the most frequently used topologies in CSA are compared. Afterwards, a frequency analysis of the amplification topology is presented and later the influence of the Miller effect is calculated. In the end, the

white noise of CSA is analyzed.

In the fourth chapter, all the topologies of Schematic design are explained, and the simulations of the circuit, which are extracted via Cadence Spectre, are presented and discussed. In the fifth chapter, the layout is presented, and simulations based on back-end-of-line parasitics are extracted, which are compared with those of chapter four.

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Chapter 1

Introduction to detectors and the Charge Sensitive Preamplifier

The present thesis concerns the design and study of a Charge Sensitive Preamplifier (CSA), an integrated circuit for the detection of particles and radiation in High Energy Physics experiments (HEP). The CSA is used in the initial stages of front-end electronics, which are readout circuits responsible for receiving and processing signals coming from detectors. In particular, signal processing corresponds to shaping the signals, storing information, and converting analog to digital information. The purpose of the CSA is to interconnect the sensor with the readout circuits.

In addition, it should be mentioned that the need to design more efficient analog systems is undeniably grave, since signals originating from physical sources are analog. In the following paragraphs, the fields of application of readout circuits, the types of sensors, and the operation of the CSA will be presented, mentioning its main characteristics. A more extensive analysis will be provided for strip and pixel detectors, as they are the most widely used in High Energy Physics experiments.

1.1 Applications of readout circuits

Readout circuits are used in many fields, both in research and industry. One of the applications of readout circuits is in high energy physics experiments in which products of some reactions are detected, such as radiation of different frequencies (photons of different energies) or other particles. These products then interact with the detector, which generates the signal that will later be processed by the readout circuits.[1][2][3][4][5].

A further use of readout circuits is in imaging applications. In fact, as previously mentioned, readout circuits can convert an analog signal produced by a detector to a digital signal, and the generated digital signal can contain information that illustrates an image. Typical examples of imaging applications are in medicine, such as X-rays, computed tomography (CT), magnetic resonance imaging (MRI), ultrasound, and endoscopy.[6][7]

In addition, these systems are used in space applications. To be more specific, spacecraft, satellites and ground-based telescopes are equipped with sensors, and therefore readout electronics are mandatory for the initial processing of signals that the sensors receive ([8][9]). Furthermore, readout electronics are used in the industrial sector at many stages of production, as well as in any device equipped with sensors. In conclusion, the need for readout electronics in sectors that utilize sensors is apparent.

1.2 Detector characteristics

Integrated detectors have been successfully used for decades in elementary particle experiments. The most common types of detectors are:

- Microstrip Detectors [10]
- Charge Coupled Devices (CCD) [11]
- Silicon Drift Detectors [12]

- Pixel Detectors [10]

Since microstrip and pixel detectors are depicted in many applications, a detailed analysis is provided for these particular types. Firstly, both types of detector consist of elements (pn diodes) connected in parallel on a semiconductor substrate. When a particle or a photon traverses the detector, a charge is generated and collected by these elements, creating a current pulse. The detector type and the placement of elements on the substrate affect the detector's resolution and efficiency, as well as electrical parameters such as input capacitance or leakage current.

It has to be highlighted that, in readout systems, the processing of the detector's signal is not limited to converting it into a digital one, but it can also provide information about the energy deposited in the detector and the timing of the event. [13]

1.2.1 Detector materials

The selection of the materials, which can be used in detector manufacturing, is based on some significant requirements. To be more exact, [13]:

- The material should provide a high signal when the energy from a particle or a photon is deposited. In order to achieve this, a small energy gap between the conduction band and the valence band of the material is required. The smaller the energy gap, the easier it will be for an electron to move from the valence band to the conduction band, therefore creating an electron-hole pair.
- The atoms of the material should have a low atomic number in experiments that focus on the detection of the particle's trajectory within the detector. The smaller the atomic number of the atom, the weaker the scattering phenomena will be, which results in accurately measuring the particle's position.
- In contrast, the atoms of the material should have a high atomic number in experiments that focus on spectroscopy (the study of the electromagnetic spectrum) and imaging applications. By increasing the atomic number, the

absorption of the incident photon's energy increases, and as a consequence, more photons are absorbed from a radiation beam.

- The material should allow high mobility of electron-hole pairs and minimize charge trapping phenomena in order to collect the maximum charge possible in a short period of time.
- The collected charge should be as large as possible. Therefore, the time for recombination of electron-hole pairs should be long.
- The leakage current should be as small as possible. This sets a low boundary for the energy gap, in order to avoid generating current due to thermal effects.
- The material should be radiation resistant, depending on the case.

The table below (1.1) presents the parameters of certain materials, which are taken into consideration when it comes to detector manufacturing [13].

Table 1.1: Important parameters of the materials used for detector manufacturing. Table from source [13]

Parameters	Si	Ge	GaAs	CdTe	CdZnTe	Diamond
Atomic number Z	14	32	31/33	48/52	48/30/52	6
Energy gap [eV]	1.12	0.67	1.43	1.44	~1.6	5.48
Density [g/cm ³]	2.3	5.3	5.4	6.1	5.8	3.5
Energy required for the electron- hole pair gener- ation [eV]	3.64	2.96	4.2	4.43	~4.6	13.1
Mobility at 300K [cm ² /Vs] -electrons -holes	1350 480	1900 3900	8000 400	1100 100	~1000 ~100	1800 1200
Recombination time of carriers [us]	~250	250	0.001-0.01	~0.1-2	~0.1-2	0.001

Silicon (Si) is the most commonly used material for detector manufacturing, as it meets all the above requirements except for the high atomic number and radiation resistance. It is typically used in particle experiments rather than in spectroscopy,

where its use is limited to low-energy photons, for which its low atomic number is acceptable.

Germanium (Ge) is the most common material for spectroscopy experiments because it has a high atomic number, which offers greater energy resolution. However, its energy gap is smaller than that of silicon, resulting in a higher leakage current, and therefore germanium detectors usually operate at low temperatures. The reason for operating in these conditions comes from the fact that when the temperature of the material drops, it is harder for the electrons to move to the conduction band.

Further materials that fit the requirements are:

- Gallium Arsenide (GaAs), which exhibits high radiation resistance and high carrier mobility. However, the electron-hole pair's recombination time is short due to impurities, making it susceptible to charge collection issues.
- Cadmium telluride (CdTe) and Cadmium zinc telluride (CdZnTe) have a high atomic number, making them suitable for spectroscopy experiments. However, the carrier mobility (specifically that of holes) is low, and integration of the chips with readout electronics is challenging.
- Diamond has a low atomic number and radiation resistance. Although it could be used in particle experiments, it requires significantly more energy to produce an electron-hole pair, since the energy gap between the conduction band and the valence band is large.

1.2.2 PN Junction

In the detector between each element (strip or Pixel) and the substrate, pn junctions are formed. Consequently, a depletion region is formed in which there are no carriers (electrons or holes) but only ions attached to the crystal lattice. The depletion region of p-type semiconductor consist of positive ions, while that of n-type consist of negative ions, therefore a potential difference is formed in the junction and as a result an electric field is created. By applying a positive potential to

the substrate (n-type semiconductor) and grounding each element, the depth of the depletion region increases, because the potential difference of the junction increases (and by extension the electric field) and consequently a greater amount of energy is required by the charge carriers to cross the junction.[14][15]

The enhancement of the electric field is of great significance because the strong electric field contributes to the collection of charge in a short period of time when electron-hole pairs are created in the detector because of radiation or particle penetration. Outside the depletion region, the electric field is weaker, and the majority carriers are in high concentrations, resulting in both poor collection of charge and high possibility of electron-hole recombination.

Having taken all the above analysis into consideration, it is evident that a wide depletion region is preferable. After calculations [13] and considering that the doping density of the p-type semiconductor (N_A) is much smaller than that of the n-type semiconductor (N_D), the thickness of the depletion region is given by the equation:

$$W_{dep} = \sqrt{\frac{2\varepsilon * (V_{bi} + V_R)}{qN_D}} \quad (1.1)$$

For which:

- ε : The permittivity of the material.
- V_{bi} : The potential difference of pn junction without applying an external potential.
- V_R : The external potential implemented.
- q : The absolute value of the electron charge.
- N_D : The doping density of an n-type semiconductor.

By solving equation (1.1) with respect to voltage, the potential that must be applied

to create a depletion region of thickness d is:

$$V_{dep} = V_R|_{W_{dep}=d} = \frac{qN_D d^2}{2\varepsilon} - V_{bi} \quad (1.2)$$

1.2.3 Detector Capacitance

The depletion region is free of charge carriers, as previously mentioned, which results in a formation of a capacitance within the detector. This capacitance is calculated [13]:

$$C = \frac{dQ_C}{dV_R} = \frac{d(qN_D W_{dep})}{dW_{dep}} * \frac{dW_{dep}}{dV_R} \Rightarrow$$
$$C = \sqrt{\frac{q\varepsilon N_D}{2(V_{bi} + V_R)}} \quad (1.3)$$

1.2.4 Leakage Current

Outside the depletion region, the majority carriers of the p-type semiconductor and those of the n-type semiconductor cannot cross the pn junction due to the large potential barrier. In contrast, the electric field of the pn junction favors the transfer of the minority carriers of each semiconductor, therefore creating a small current in the detector. This current is called leakage current or thermally generated reverse current. The reason it is called thermal is that the minority carriers are generated by the breaking of covalent bonds due to the increase in temperature. As a result, a way of reducing the leakage current is to lower the temperature of the detector. [13]

1.3 Single-sided microstrip detectors

Microstrip detectors consist of pn diodes connected in parallel. Specifically, the substrate is made of an n-type semiconductor, while strips of a p-type semiconductor are placed on top of it. Next, an insulating material such as silicon dioxide (SiO_2) is placed above the detector, and on top of that, in the same positions where the p-type elements were placed, metal strips are placed. The bottom side of the substrate is

the ohmic contact of the detector and is metallized, in the case of a single-sided microstrip detector (Figure 1.1). [13][16]

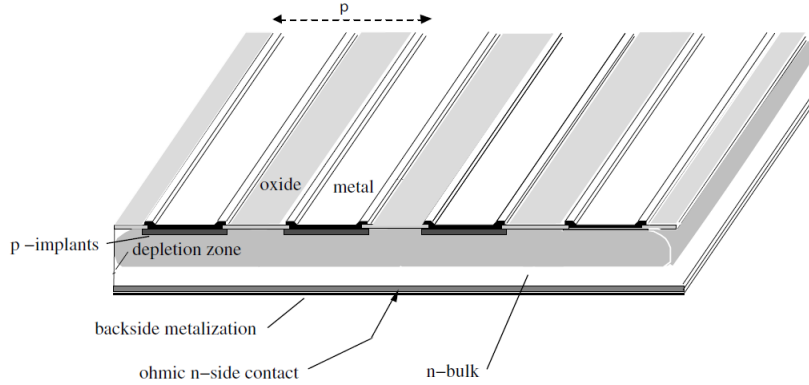


Figure 1.1: Single-sided microstrip detector of p-type strips. The strips are placed in p distance with each other. The figure comes from [16] source.

1.3.1 Double-sided microstrip detectors

In the previously described setup, detection was possible in only one dimension. However, microstrip detectors capable of detecting in two dimensions have also been designed. This type of microstrip detector has groups of p-type semiconductor strips both on the top and bottom, which are rotated 90 degrees relative to each other. However, in the case of simultaneous event detection (two particles or two photons interacting with the material at the same time), both true and false interaction points are created (Figure 1.2b). According to the figure, two events were detected, but they correspond to four positions. [13]

1.4 Pixel Detector

The structure of the pixel detector is similar to that of the microstrip detector, with the difference that the semiconductor strips placed on the substrate are subdivided into smaller pieces (pixels). If the surface area of the pixels is larger than a few square millimeters and the number of pixels is about 100, then the readout electronics can be connected to the edges of the detector (pad detector), as shown in 1.3a. However, if their dimensions are smaller than 1 mm^2 and the number of pixels is

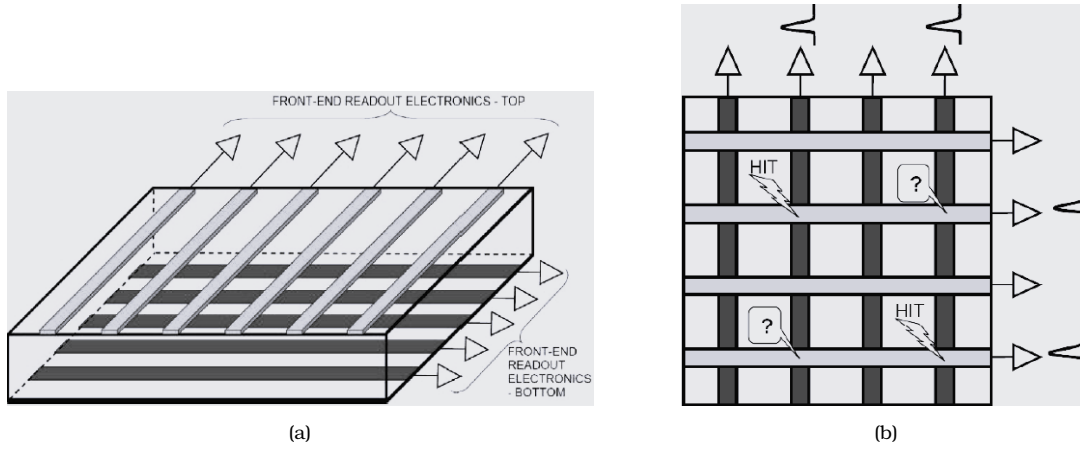


Figure 1.2: (a) Microstrip detector with two groups of parallel strips, capable of detecting in two dimensions. (b) Incorrect generation of interaction positions when more than one photon or particle interacts with the detector material simultaneously. Diagram from source [13].

between 10^3 and 10^4 , then a two-dimensional connection of the readout electronics to the pixel detector is necessary (**1.3b**). The connection between the detector and the readout electronics should be made vertically to the base of the sensor, not at the edges, as in pad detectors. Furthermore, the dimensions of the pixel should be compatible with those of the readout electronics, and the readout integrated circuit should be placed very close to the sensor, at a distance of 10-20 μm . [16]

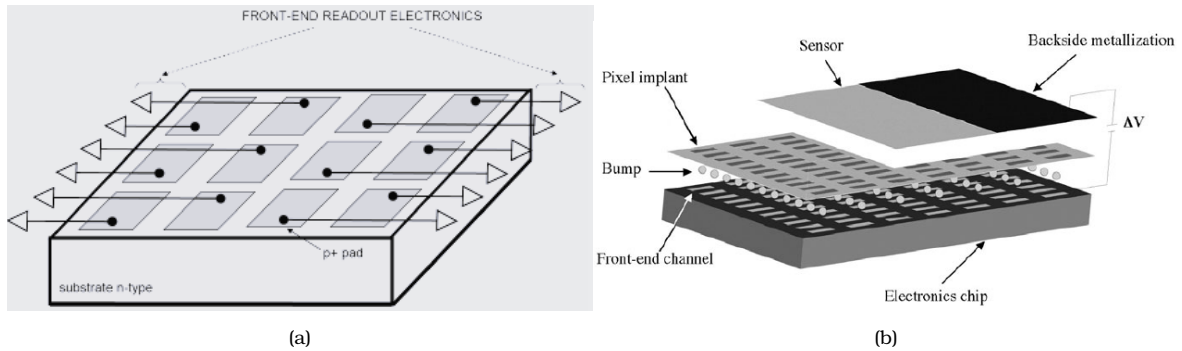


Figure 1.3: (a) Pad detector. (b) Pixel Detector. Diagram from source [13] and [16].

Similarly to the microstrip detector, the parallel pn diodes are reverse biased. To achieve this, the top side, which forms the ohmic contact of the detector substrate, is connected to a high potential, while each pixel is grounded through the preamplifier circuit. This type of detector exhibits low capacitance and low leakage current due to the small size of the pn diodes. [16]).

1.5 Charge Generation inside the Detector

The incident photon or particle, in both types of detectors, interacts with the substrate material and produces electron-hole pairs. The interaction mechanism varies depending on the energy of the incident particle or photon.

In the case of electromagnetic radiation, the photon interacts with the material at a certain point in the substrate and creates an electron-hole pair near that point. The mechanisms by which the photon can interact with the material and produce an electron-hole pair include the photoelectric effect, the Compton effect, and pair production. Ultimately, the number of electron-hole pairs generated by these phenomena is determined by the equation:

$$N = \frac{E_{\text{photon}}}{E_{\text{electron-holes}}} \quad (1.4)$$

In which E_{photon} is the energy of the photon, and $E_{\text{electron-holes}}$ is the energy required to form an electron-hole pair. It is noteworthy that the energy required for the formation of the pair has some variation, as the boundaries of the conduction band and that of the valence band in a semiconductor are not clearly defined. Therefore, the number of pairs exhibits some fluctuation.

In the case of particle interaction with the substrate, electron-hole pairs are created along the entire path of the particle through the substrate. The same equation (1.4) expresses the number of pairs created. Similarly to the case of photons, the number of pairs produced exhibits fluctuation for the same reason. Figure (1.4) shows the ways in which photons and particles interact with matter.

The generated charge produced by the interaction of the particle or photon with the matter is transported within the matter, either by drift or diffusion. Specifically:

- Drift: The charge drifts due to the electric field present in the depletion region and is scattered along its path by the atoms of the crystal lattice.
- Diffusion: The generated charge carriers can diffuse into neighboring areas due

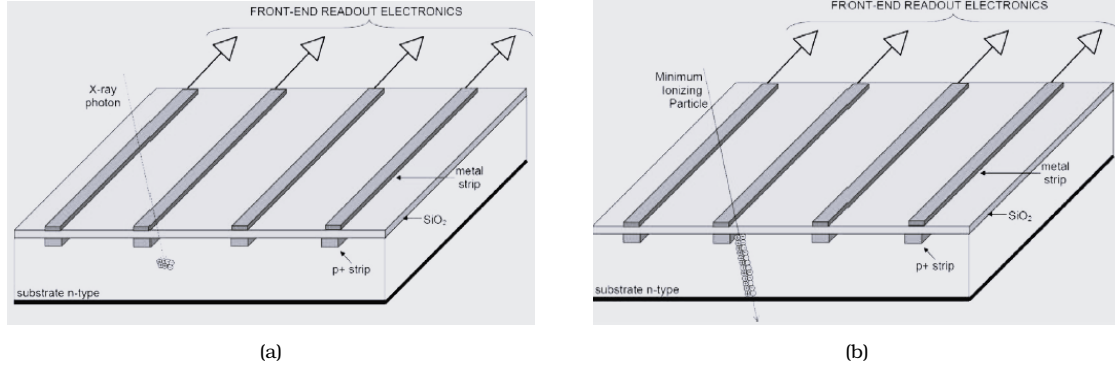


Figure 1.4: Production of electron-hole pairs in the case of: (a) photon interaction with matter and (b) particle interaction with matter. Diagram from source [13].

to the lower carrier concentration that may exist in those regions.

It is worth noting that due to diffusion, the charge may be spread across neighboring strips (microstrip detectors) or neighboring pixels (pixel detectors), causing detection problems. [13]

1.6 Charge Transport

The time required for the charges to traverse the detector is called collection time and is considered an important characteristic of the detector. The collection time depends on the interaction point, the thickness of the detector, and the electric field. Assuming that the applied voltage is higher than V_{dep} (in order for the depletion region to cover the entire thickness of the detector), the collection times (detailed calculations are available in source [13]) for electrons and holes are respectively:

$$t_e = \frac{\epsilon}{\mu_e q N_D} \ln \frac{x_0 + \frac{\epsilon}{q N_D} E_{min}}{\frac{\epsilon}{q N_D} E_{min}} \quad (1.5)$$

$$t_h = \frac{\epsilon}{\mu_h q N_D} \ln \frac{d + \frac{\epsilon}{q N_D} E_{min}}{x_0 + \frac{\epsilon}{q N_D} E_{min}} \quad (1.6)$$

For which:

- μ_e and μ_h : They are the mobility of electrons and holes, respectively.
- x_0 : It is the distance from the interaction point measured from the depletion

region with the highest potential.

$$- E_{min} \approx \frac{V_R - V_{dep}}{d}.$$

1.7 The Charge Sensitive Preamplifier

The Charge Sensitive Preamplifier [CSA] is used for the interfacing of the detector with the readout electronics. The current pulse from the detector is integrated and amplified by the preamplifier, producing a voltage pulse at its output. The simplified circuit of the CSA is depicted in the figure.(1.5a).

Furthermore, the number of CSAs and consequently readout electronics required for a detector depends either on the number of strips in the case of microstrip detectors or the number of pixels in the case of pixel detectors. The readout electronics are designed in groups of 32, 64, and 128 channels. A typical example is illustrated in figure 1.5b). [13]

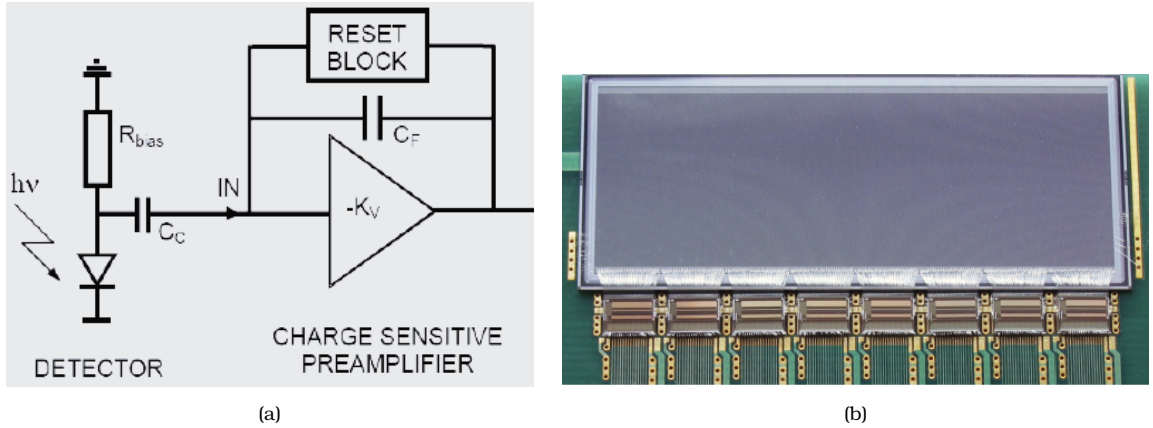


Figure 1.5: (a) Simplified circuit of the CSA (diagram from source [13]). (b) Microstrip detector with 512 strips connected to 8 groups of 64-channel readout electronics (diagram from source [13].)

1.7.1 Response time of Charge Sensitive Preamplifier

In figure (1.6) a more detailed view of the CSA circuit is depicted. In particular:

- The feedback resistor is connected in parallel with the feedback capacitor to

discharge the capacitor when there is no current pulse in order to prevent the output to reach saturation.

- The detector is modeled as a current source in parallel with a capacitor. The current pulse that the detector produces is considered a Dirac delta current pulse, which the integral of this pulse is equal to the charge of an electron at the input ($i_D(t) = Q_{in}\delta(t)$).
- The amplification stage of the CSA is considered to have a single pole, which is the dominant one (in reality there are more (Chapter 3)).

The transfer function of the amplification stage of the CSA is given by the equation (source [13]):

$$\frac{V_{out}(s)}{V_{in}(s)} = K = -\frac{K_0}{1 + \frac{s}{\omega_0}} \quad (1.7)$$

For the above expression, K_0 is the DC gain and ω_0 is the dominant pole of the amplification stage. The sign is negative because, in the Folded Cascode topology which is used, the gain is negative. Therefore, the gain-bandwidth product is:

$$GBW = K_0\omega_0 \quad (1.8)$$

By examining the figure (1.6), the time and frequency response of the CSA can be determined. It is noteworthy that a buffer is connected in series with the preamplifier to allow the connection of large loads at the output. From the input node of the illustrated circuit in figure (1.6) it can be extracted:

$$i_D = V_{in}sC_T + (V_{in} - V_{out})\left(\frac{1}{R_f} + sC_F\right) \quad (1.9)$$

Substituting equation (1.7) into equation (1.9) yields the transfer function.

$$\frac{V_{out}(s)}{i_D} = -\frac{K_0}{\frac{1+K_0}{R_f} + s[C_T + (1 + K_0)C_F + \frac{1}{\omega_0 R_f}] + s^2 \frac{C_T + C_F}{\omega_0}} \quad (1.10)$$

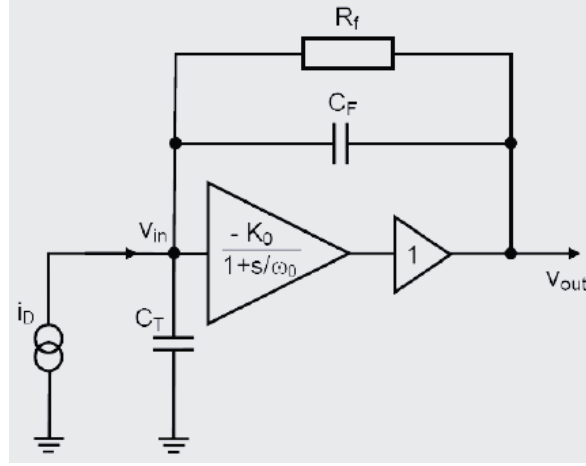


Figure 1.6: Detailed presentation of the CSA (diagram from source [13]).

Assuming that $K_0 \gg 1$, $(K_0 + 1)C_F \gg C_T$ and multiplying the numerator and denominator of equation (1.10) with $\frac{R_f}{K_0}$, it results in:

$$\frac{V_{out}(s)}{i_D(s)} \approx -\frac{R_f}{1 + s[R_f C_F + \frac{1}{GBW}] + s^2 \frac{(C_T + C_F)R_f}{GBW}} \quad (1.11)$$

From the above equation (1.11), it is apparent that the transfer function has two poles. Additionally the denominator of 1.11 can be expressed as ($\omega_1 \ll \omega_2$):

$$D(s) = (1 + \frac{s}{\omega_1})(1 + \frac{s}{\omega_2}) = 1 + s(\frac{1}{\omega_1} + \frac{1}{\omega_2}) + \frac{s^2}{\omega_1 \omega_2} \approx 1 + \frac{s}{\omega_1} + \frac{s^2}{\omega_1 \omega_2} \quad (1.12)$$

From equations 1.11 and 1.12, and for $R_f C_F \gg \frac{1}{GBW}$ the dominant pole and the second pole are obtained.

$$\omega_1 \approx \frac{1}{(R_f C_F)} \quad (1.13)$$

$$\omega_2 \approx GBW \frac{C_F}{C_T + C_F} \quad (1.14)$$

As a consequence, there are two time constants ($\tau_f = 1/\omega_1$ and $\tau_2 = 1/\omega_2$), which are associated with the above poles. Substituting the time constants into equation 1.11 yields:

$$\frac{V_{out}(s)}{i_D(s)} \approx -\frac{R_f}{(1 + s\tau_f)(1 + s\tau_2)} = -\frac{1}{C_F} \frac{\tau_f}{\tau_f - \tau_2} \left(\frac{\tau_f}{1 + s\tau_f} - \frac{\tau_2}{1 + s\tau_2} \right) \quad (1.15)$$

1.7. The Charge Sensitive Preamplifier

Applying the Laplace transform and considering a Dirac delta current pulse as the input, the time response of the CSA is obtained.

$$V_{out}(t) = -\frac{Q_{in}}{C_F} \frac{\tau_f}{\tau_f - \tau_2} [\exp(-t/\tau_f) - \exp(-t/\tau_2)] \quad (1.16)$$

The time constant τ_f is responsible for the exponential decay of the output signal, while the constant τ_2 is responsible for the exponential rise. Therefore, the output waveform will be the linear combination of the waveforms in figures (1.7a) and (1.7b), respectively.

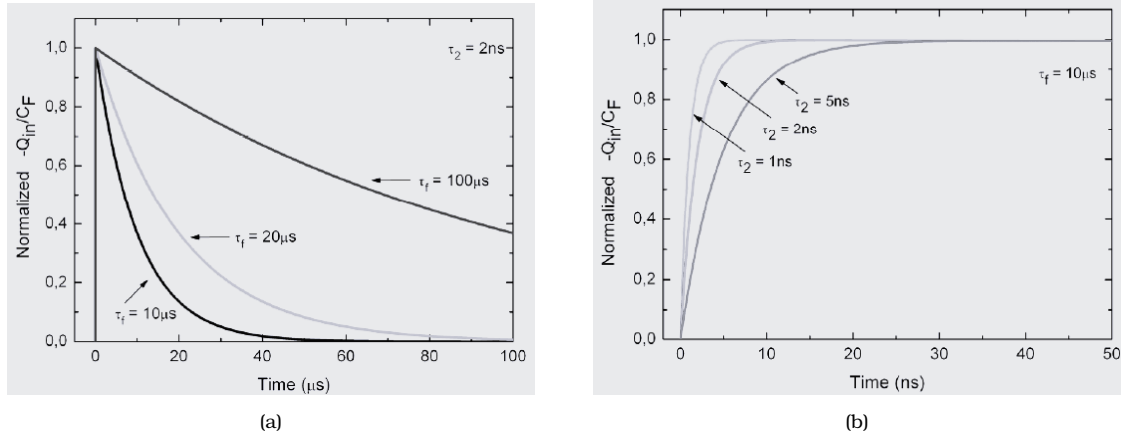


Figure 1.7: (a) Exponential decay of the output waveform for $\tau_2 = 0$. (b) Sharp exponential rise of the output waveform for $\tau_f = 0$ (Diagrams from source [13]).

Chapter 2

Types of noise

2.1 Noise Definition

Noise, in the broader sense, is any unwanted disturbance that interferes with the desired signal, creating problems in the transmission, processing, and analysing of the signal. Depending on the noise source, the noise can be characterised as external or internal noise.

Firstly, external noise is caused by external sources, such as interference from neighboring circuits, vibrations, or radiation. When noise is external, it can be reduced or even eliminated without the need to change the circuit layout design.

On the other hand, internal noise is caused by the components of the circuit and the materials by which they are constructed. Therefore, it can be considered as a small internally generated signal of the device. Unfortunately, this type of noise cannot be eliminated, but can only be reduced by making appropriate design choices. As a consequence, internal noise defines the minimum size of the signal. This chapter is dedicated exclusively to internal noise. [17][18][19]

2.2 Mathematical approach

The source-drain current of a MOSFET transistor consists of an ideal DC current and a time-varying current, which is generated due to the internal noise of the device.

$$i_{DS}(t) = I_{DS} + i_n(t) \quad (2.1)$$

The representation of a MOSFET's source-drain current (equation 2.1) is depicted in figure (2.1b). The current is defined as the movement of charge particles. Since

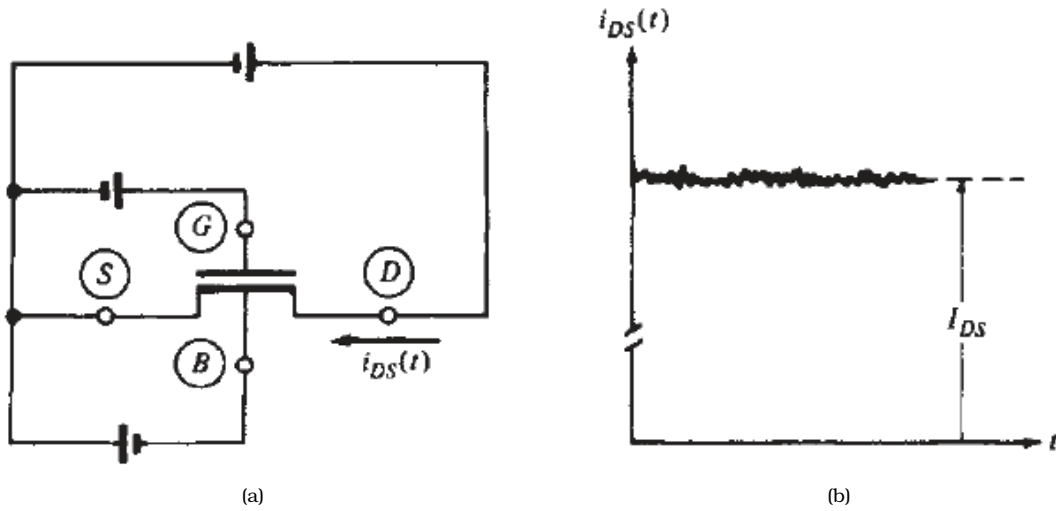


Figure 2.1: Representation of a MOSFET's source-drain current. Figure from source [17].

charge particles contain a certain amount of charge, the current is quantized and not continuous. The discontinuity of the electric current primarily causes the noise; however, the average value of the electric current is considered as continuous. Therefore, to quantify the noise, average values of quantities have to be used, specifically the mean square value of the noise current or voltage and their Root Mean Square (RMS).

Another quantity that is used to quantify the noise is the Power Spectral Density (PSD). A common measurement of noise voltage or current is performed over a very narrow frequency range. Dividing the mean square value of the current or voltage by this infinitesimal frequency range, the Power Spectral Density is obtained. Equations (2.2) and (2.3) show the mean square value and the Root Mean Square value of the

noise current, respectively [17]:

$$\overline{i_n^2} = \int_{f_1}^{f_2} S_i(f) df \quad (2.2)$$

$$i_{n,rms} \sqrt{\overline{i_n^2}} = \sqrt{\int_{f_1}^{f_2} S_i(f) df} \quad (2.3)$$

The two main categories of noise are:

- **White noise:** It is the type of noise where the power spectral density remains constant across the entire frequency spectrum. The sources of white noise are well-known, and its intensity can be predicted.
- **Low frequency noise:** At low frequencies, flicker noise predominates. This noise increases inversely with frequency. There are various theories about its origin in CMOS technology, with two of them being the most prevalent.

In a noise-frequency line graph, the frequency spectrums in which each noise type predominates are separated by a corner frequency (f_c). This frequency ranges from a few Hertz to Megahertz depending on the component's construction, geometry, and biasing. The contributions of white noise and $1/f$ noise to the total noise are independent of each other, so the mean square value of the total noise can be expressed as:

$$\overline{i_n^2} = \overline{i_w^2} + \overline{i_f^2} \quad (2.4)$$

Dividing equation (2.4) by an infinitesimal frequency range yields the relationship of noise Power Spectral Density:

$$S_i(f) = S_{iw}(f) + S_{if}(f) \quad (2.5)$$

A typical diagram for the power spectral density of a MOSFET's drain-source noise current is illustrated in figure (2.2):

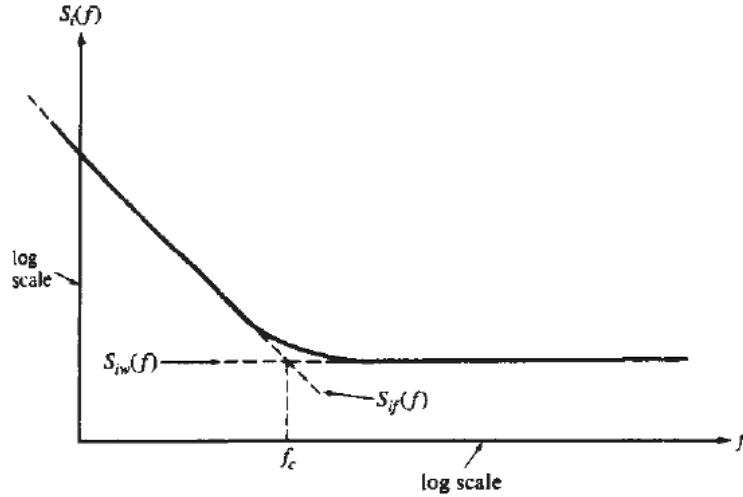


Figure 2.2: PSD of a MOSFET's drain-source noise current. The corner frequency is apparent, beyond which the noise remains constant. Figure from source [17]

2.3 Noise in CMOS technology

In CMOS technology, the dominant sources of noise are thermal noise (specifically, the thermal noise of the channel) and $1/f$ noise. The following sections will analyse these two types of noise, as well as the shot noise.

2.3.1 Thermal Noise

Thermal noise is caused by the random thermal motion (vibration) of charge carriers in any conductor that is at a temperature above absolute zero, similar to the Brownian motion of particles. It has been found that the noise power P_n is proportional to the absolute temperature and the frequency range:

$$P_n = kT\Delta f \quad (2.6)$$

for which $k = 1.38 \times 10^{-23} \text{ J/K}$ is the Boltzmann's constant. In addition, from equation (2.6), the RMS value of noise voltage can be derived [18]:

$$v_n = \sqrt{4RkT\Delta f} \quad (2.7)$$

If square the equation (2.7) and divide it by the frequency range yields the power spectral density.

$$S_v = 4kTR \quad (2.8)$$

2.3.1.1 Channel Thermal Noise

Regarding channel thermal noise, the source of this noise is also due to the random thermal motion of carriers in the transistor's channel. When V_{GS} is above the threshold voltage and $V_{DS} = 0$, the channel depth is uniform across its extent. Therefore, it can be considered as a resistor, and the equations that were developed above can satisfactorily describe the thermal noise in the channel. Specifically, from equation (2.7), the noise current in the channel is given by [21]:

$$i_n^2 = 4kTg_m \quad (2.9)$$

For which g_m is the channel transconductance.

However, the above discussion represents an ideal case. In most applications, there is a potential difference between the drain and the source of the MOSFET device, causing the channel depth to vary across the channel length. For example, in the case of an NMOS transistor, the channel narrows towards the drain, because the potential in the drain node is greater than that in the source. As a consequence, the channel depth near the drain will be smaller than that near the source. In order to find the square value of the channel current, it is necessary to integrate over the varying potential in the channel:

$$i_n^2 = 4kT \frac{\mu^2 W^2}{L^2 I_{DS}} \int_0^{V_{DS}} \mathcal{Q}_n^2(V) dV \quad (2.10)$$

For which:

- W and L : The dimensions of the channel.
- μ : The mobility of the charge carriers inside the channel.

- I_{DS} and V_{DS} : The DC component of the channel current and the potential difference between the source and the drain.
- $Q_n(x) = C_{ox}(V_{GS} - V_T(x) - V(x))$
 - C_{ox} : The oxide capacitance.
 - $V_T(x)$: The threshold voltage.
 - $V(x)$: The voltage inside the channel at a particular point.

From equation (2.10), the source-drain current of the MOSFET for the linear region, the saturation region, and the boundary between the two, can be derived. In the linear region, the resulting equation is the same as equation (2.9). For the boundary between the linear region and the saturation region, the resulting equation is:

$$i_n^2 = \frac{8}{3}kTg_m \quad (2.11)$$

Additionally, it has been experimentally found that in the saturation region, the current can be satisfactorily approximated by equation (2.11) (it has to be mentioned that the body effect has not been included in the previous calculations). In reality, the threshold voltage depends on the varying potential in the channel, and the general form of equation (2.11) becomes:

$$i_n^2 = 4kT\gamma g_m \quad (2.12)$$

For which γ is a function of the transistor parameters.

2.3.2 Flicker Noise

Flicker noise is observed in many types of devices such as resistors, diodes, thermistors, transistors, and others. The frequency spectrum of flicker noise is expressed by the function $1/f^\alpha$, where α is close to one. There is no absolute theory that describes the origin of this noise, however; for CMOS devices, the two dominant

theories regarding the generation of flicker noise are the Hooge and McWhorter theory.

2.3.2.1 Hooge Theory

Hooge's theory is based on the fluctuation in carrier mobility during their movement in the transistor's substrate and is known as the Mobility Fluctuation Model. In this analysis, it was assumed that the construction materials are homogeneous; otherwise, differential equations would be necessary to find the noise current. Hooge's equation is empirical, which according to the previous assumption is [21]:

$$\frac{i_f^2}{I^2} = \frac{a_i}{N * f} \quad (2.13)$$

For which:

- N: The number of charge carriers.
- a_i : The Hooge parameter.
- I: The short-circuit current of the transistor.

It should be noted that the results obtained by this model to the linear and saturation region, have exhibited deviations and different dependency relationships in comparison to some experimental results.

2.3.2.2 McWhorter Theory

The theory of McWhorter, also known as the Number Fluctuation Model, suggests that flicker noise arises from the random trapping and release of charge carriers at the silicon junction of the channel with the silicon dioxide $Si - SiO_2$ and within the gate oxide. The combination of these two sources are considered by the McWhorter theory as the cause of flicker noise.

Combining the two models, the result in the root mean square value of the noise

voltage for the saturation region is ([21]):

$$\overline{v_f^2}(f) = \frac{K_f}{C_{ox}^2 WL * f} \quad (2.14)$$

The parameter K_f in the above equation takes different values for *NMOS* and *PMOS* transistors. Its value also depends on the technology that will be used, the bias point of the transistor, and whether it is operating in the subthreshold region or not. Equation (2.14) is considered a good but simplified approximation of the flicker noise. [20][21]

2.3.3 Shot Noise

Shot noise occurs in semiconductor contacts, such as diodes and transistors, as well as in vacuum tubes. The contact current equals to the sum of the charges passing through the contact per unit of time, and thus it is not continuous. When an electron crosses the pn junction, a current pulse is generated, resulting in small fluctuations. These small fluctuations caused by the charge carriers are called shot noise, and the RMS value of the noise current is given by the relationship [18]:

$$I_{sh} = \sqrt{2qI_{DC}\Delta f} \quad (2.15)$$

For Which:

- q : It is the absolute value of electron charge ($1.602 * 10^{-19}$).
- I_{DC} : It is the DC current of the junction.
- Δf : It is the noise frequency spectrum.

From equation (2.15), the power spectral density is found:

$$S_{sh} = 2qI_{DC} \quad (2.16)$$

From the power spectral density formula, it is observed that PSD remains constant

across the entire frequency range, therefore shot noise corresponds to white noise. The existence of current is necessary for the generation of shot noise. In a pn junction, there is current due to the majority carriers and current due to the minority carriers.

In addition, shot noise is produced at the pn junction when it is forward biased, since current flows through the pn junction and therefore shot noise cannot be eliminated in this case. Furthermore, when the junction is not biased or is reverse biased, shot noise is still produced despite the absence of current through the junction. This occurs because the current of the minority carriers always exists, and it cannot be eliminated. For this reason, the pn junction is modelled as a noiseless dynamic resistance connected in parallel with a noise current source.

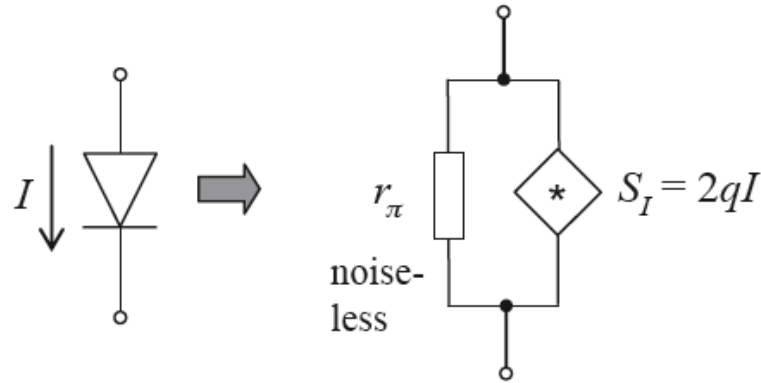


Figure 2.3: Modelling of the pn junction as a parallel combination of a noiseless dynamic resistance and a noise current source. Figure from source [3].

Chapter 3

Analysis of the amplification stage of the Charge Sensitive Preamplifier

The reason the Charge Sensitive Preamplifier is a common choice when it comes to particle detection experiments is because it has high open-loop gain, it collects adequately the charge generated by the detector, and it minimizes interference between channels (Crosstalk). The analysis in this chapter is based on the published reference [1].

3.1 Amplification topologies of the Charge Sensitive Preamplifier

For the circuitry of the CSA, the most common amplification topologies are the cascade (connecting common-source amplifiers in series) and the cascode topology (connecting the drain of a common-source amplifier to the source of a common-gate amplifier). In the cascode topology, the desired gain is achieved by increasing the output resistance, whereas in the cascade topology, at least three amplification stages are designed for this purpose. The cascode topology has prevailed over the cascade topology for three main reasons.[1]

- The Miller effect is limited. Miller's theorem states that when a complex impedance connects two nodes of a circuit, it can be replaced by two complex impedances connecting the nodes to ground. By applying Miller's theorem to

a bridging capacitor, a capacitor is formed at the input with a value equal to the bridging capacitor multiplied by the factor $1-K$ (where K is the gain of the amplifier). This multiplication is referred to as the Miller effect.[15]

- It has a dominant low-frequency pole, resulting in behavior similar to that of a single-stage amplifier.
- The in-series contribution of noise is limited only to the input transistor. Therefore, the signal-to-noise ratio (SNR) is higher.

3.2 The effect of the transistor shrinking on the CSA

A metric used to characterize how faster the response of a circuit becomes due to the reduction in transistor dimensions is the transit frequency of a MOSFET (f_T). This is the frequency at which the AC current of the MOSFET's gate becomes equal to the AC current in its drain. The formula for the transit frequency for a transistor biased with a current I_{SPEC} is [1]:

$$f_T \cong \frac{1.15K_p t_{ox} U_T}{\pi \epsilon_{ox} L^2} = \frac{1.15\mu_0 U_T}{\pi L^2} \quad (3.1)$$

For which:

- K_p : The conductance parameter.
- t_{ox} and ϵ_{ox} : They are the thickness and the electrical transmittance of the oxide in the gate area, respectively.
- U_T : It is the thermal potential.
- L : It is the length of the channel.
- μ_0 : It is the charge mobility.

The increase in the transit frequency and consequently the speed of electronics during the evolution of technology was not achieved due to a rise in the mobility of

charge carriers, but rather due to the transistor shrinking. In fact, as technology evolved, the mobility of carriers decreased, resulting in a decrease in conductivity and amplification provided by MOS elements. This condition created a greater need for amplifying devices.

3.3 Frequency response of the CSA's amplification stage

3.3.1 Folded Cascode amplifier

In the designed CSA, the topology of the Folded Cascode amplifier was used instead of the Cascode. This choice was made because its common-mode operating range is greater than that of the Cascode and the DC level of the output signal can be the same as that of the input [23]. In Figure (3.1), the topologies of the Cascode and Folded Cascode amplifiers are illustrated, while Figure (3.2) depicts the small-signal analysis of the Folded Cascode amplifier. From figure (3.2) the following equations

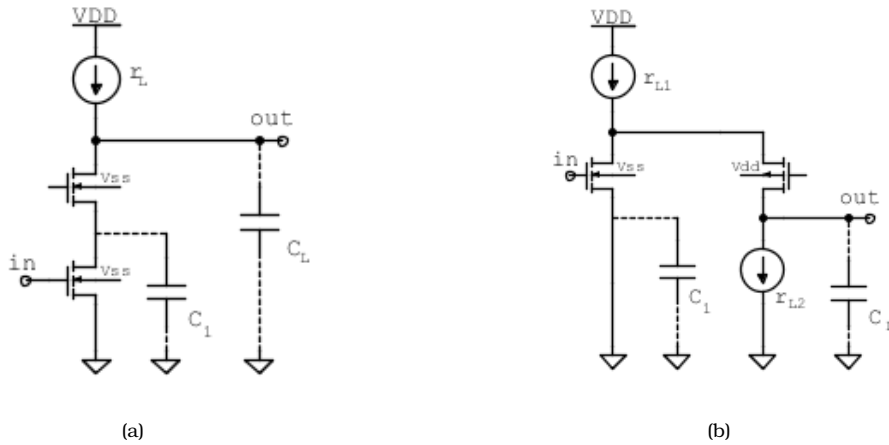


Figure 3.1: (a) Cascode amplifier implemented NMOS transistor. (b) Folded Cascode amplifier.

$$i_2 = \frac{u_0}{Z_L} - g_{m2}u_1 \quad (3.2)$$

$$i_2 = (u_1 - u_0)/(r_{ds2}) \quad (3.3)$$

can be extracted:

$$i_2 = -u_i g_{m1} - u_1 \left(\frac{1}{R_s} + g_{m2} + sC_1 \right) \quad (3.4) \quad Z_L = r_{L2}/(1 + sC_L r_L) \quad (3.5)$$

$$R_s = (r_{ds1} * r_{L1})/(r_{ds1} + r_{L1}) \quad (3.6)$$

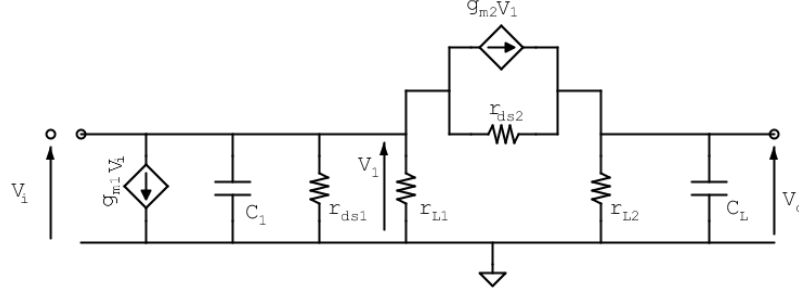


Figure 3.2: Small signal analysis of the Folded Cascode amplifier.

By solving the equations 3.2 to 3.6, the transfer function of the Folded Cascode amplifier and the first amplification stage is obtained.

$$K_u(s) = -\frac{g_{m1} r_{ds2}}{(1 + \frac{s}{P_1})(1 + \frac{s}{P_2})} \quad (3.7)$$

$$P_1 = \frac{1}{C_L r_{ds2}}, P_2 = \frac{g_{m2}}{C_1} \quad (3.8)$$

The transfer function of the first amplification stage:

$$K_{u1}(s) = -2 \frac{g_{m1}}{g_{m2}} \frac{1 + \frac{s}{Z}}{(1 + \frac{s}{P_1})(1 + \frac{s}{P_2})} \quad (3.9)$$

$$Z = \frac{2}{C_L r_{ds2}}, P_1 = \frac{1}{C_L r_{ds2}}, P_2 = \frac{g_{m2}}{C_1} \quad (3.10)$$

In order to make these calculations, three assumptions were made:

1. It was assumed that the internal resistance of the source I_{bias2} is equal to the resistance of the Cascode transistor (common-gate transistor) ($r_{L2} = r_{ds2}$), in order to simplify the calculations.
2. The inequality $g_m \gg g_{ds}$ holds because the transistor's current is affected in a greater extend by the voltage V_{gs} than by V_{ds} .
3. The poles of the amplifier are located in distant frequency regions from each other.

It is apparent that the transfer function (and therefore the gain) is proportional

to the transconductance of the input transistor and the resistance of the cascode transistor. To increase the transconductance g_{m1} , it is sufficient to increase the width of the input transistor and reduce its length. Another method is to increase the bias current, with respect to power consumption specifications. To increase resistance r_{ds2} , the cascode transistor should have long length and small width. Both methods lead to an increase in amplification.

Furthermore, from equation (3.8), it is clear that the Folded Cascode exhibits two poles. The first pole depends on the capacitance that is present in the drain of the cascode transistor and its resistance. The second pole is at a higher frequency and depends on the transconductance of the cascode transistor and the capacitance that is present at the input transistor. [1][22][15]

3.3.2 Ways of increasing the amplification of the Folded Cascode amplifier

In order to further increase the gain, a cascode topology like the one shown in Figure (3.1a) is used as the current source I_{bias2} . In this way, the load of the current source I_{bias2} is increased, which enhances the gain of the Folded Cascode amplifier.

Assuming the same premises as before and considering that the load in the drain of the cascode transistor is equal to the external resistance of the Folded Cascode amplifier ($r_{L2} = g_{m2}r_{ds2}(r_{ds1}||r_{L1})$) to simplify the equations, the following equations are formed. For the transfer function of the Folded Cascode amplifier:

$$K_u(s) = -\frac{1}{2} \frac{g_{m1}g_{m2}r_{ds1}r_{ds2}}{(1 + \frac{s}{P_1})(1 + \frac{s}{P_2})} \quad (3.11)$$

$$P_1 = \frac{2}{C_L g_{m2} r_{ds1} r_{ds2}}, P_2 = \frac{g_{m2}}{C_1} \quad (3.12)$$

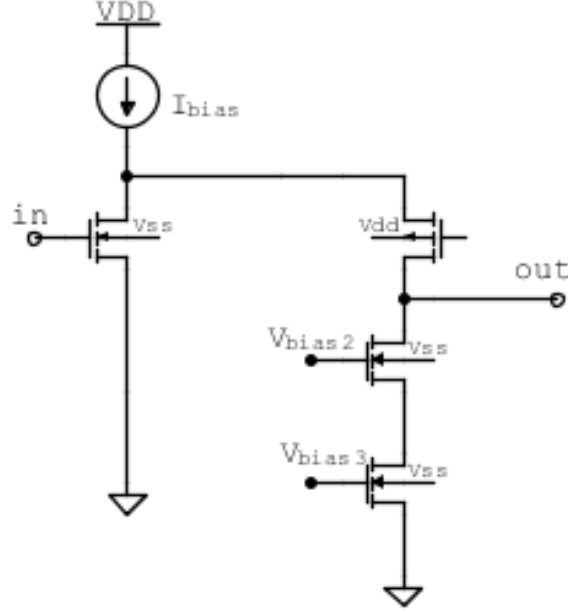


Figure 3.3: Increasing the amplification of the Folded Cascode by using a Cascode topology as current source.

For the transfer function of the first amplification stage:

$$K_{u1}(s) = -\frac{1}{2} \frac{g_{m1} r_{ds1} (1 + \frac{s}{Z})}{(1 + \frac{s}{P_1})(1 + \frac{s}{P_2})} \quad (3.13)$$

$$Z = \frac{1}{C_L r_{ds2}}, P_1 = \frac{2}{C_L g_{m2} r_{ds1} r_{ds2}}, P_2 = \frac{g_{m2}}{C_1} \quad (3.14)$$

It is apparent from the equations (3.11) and (3.12) that the amplifier's gain increases and the dominant pole shifts to lower frequencies. However, the gain-bandwidth product remains constant (GBW). [1]

3.3.3 Comparison of amplification topologies

In figure (3.4), the gain diagram is depicted for the two topologies. The diagram is for Cascode amplifiers; however, the same result is obtained for Folded Cascode amplifiers, as the frequency response formulas are identical. The Cascode LG refers to the amplifier with a transistor acting as a current source, while the Cascode HG refers to the amplifier with a Cascode topology acting as a current source.

From the above diagram it can be concluded that the cascode HG topology

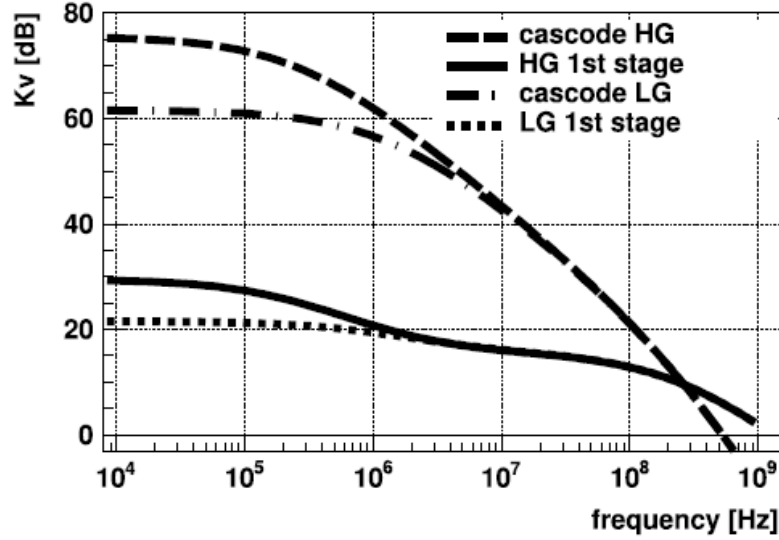


Figure 3.4: Gain chart of Cascode LG and HG amplifiers from source [1].

is preferable as it exhibits greater gain. However, in order to have a trustworthy conclusion of which topology is preferable, the Miller effect of the two topologies should also be examined. [1]

3.4 Miller effect

In this section, the Miller effect on Cascode HG and Cascode LG amplifiers will be examined and compared. In the following analysis, two assumptions were made:

1. Only the dominant pole of the CSA was taken into consideration.
2. The CSA operates only with feedback capacitance.

From the analysis of the CSA schematic (figure 3.5), the following equations were extracted:

$$u_1 = [u_i K u_1 (1 + \frac{s}{Z})] / (1 + \frac{s}{P_1}) \quad (3.15) \quad u_o = (u_1 K u_2) / (1 + \frac{s}{Z}) \quad (3.16)$$

$$i_f = s C_f (u_i + u_o) \quad (3.17) \quad i_i = u_i s (C_d + C_g) + i_f + i_{f1} \quad (3.18)$$

$$i_{f1} = s C_m (u_i + u_1) \quad (3.19) \quad K = K_{U1} K_{U2} \quad (3.20)$$

The capacitances C_d, C_g, C_m, C_f refer to the capacitances of the detector, the gate-

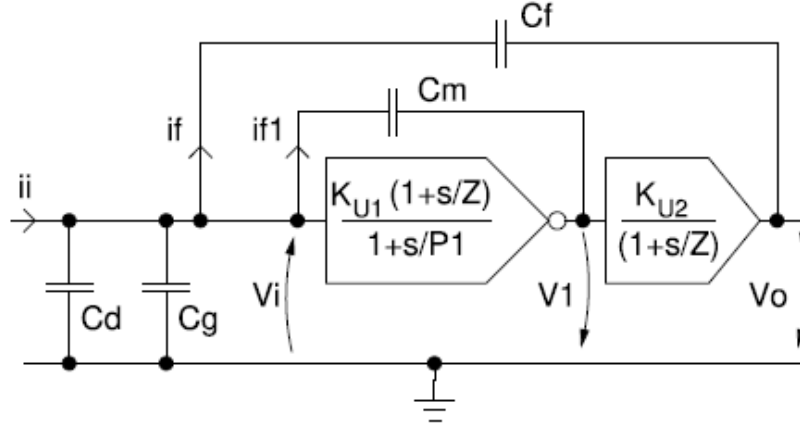


Figure 3.5: CSA schematic for studying the Miller effect on the output signal. Figure from source [1].

source of the input transistor, the gate-drain of the input transistor, and the feedback loop, respectively.

In addition, the pulse generated by the detector corresponds to a Dirac current pulse ($I(t) = q\delta(t)$). Solving for u_o the equations 3.15 to 3.20 and using the Laplace transform, the result in the time domain is:

$$u_o(t) = A(1 - \exp(-\frac{t}{\tau_p})), A \approx \frac{1}{C_f} \quad (3.21)$$

The time constant, which describes the time response of the amplifier is equal to:

$$\tau_p = \frac{C_{in}''}{P_1(C_{in}') + K_U C_f} \approx \frac{C_{in}''}{GBW C_f} \quad (3.22)$$

For which:

$$\begin{aligned} C_{in}' &= C_d + C_g + C_f + (1 + K_{U1})C_m \\ C_{in}'' &= C_d + C_g + C_f + (1 + K_{U1}P_1/Z)C_m \end{aligned} \quad (3.23)$$

It is evident that the Miller effect depends on the gain of the first stage, the dominant pole, and the zero of the first stage amplification transfer function. By substituting these values for both the Cascode LG and Cascode HG, and for $K_{U1}(0)$, it can be calculated:

$$C_{in}'' = C_d + C_g + C_f + (1 + \frac{g_{m1}}{g_{m2}})C_m \quad (3.24)$$

It is obvious that the Miller effect is the same for both types of amplifiers. In fact, the gain in capacitance for both amplifiers is described by the term g_{m1}/g_{m2} . Therefore, the cascode HG is commonly chosen as the amplification stage for the design of the CSA.

It is worth mentioning that the time constant of the amplifier is proportional to the input capacitance and inversely proportional to the gain-bandwidth product (GBW) and the feedback capacitance. Therefore, the circuit should be designed to have low input capacitance and high gain-bandwidth product. Increasing the feedback capacitance is not preferred, as it increases the power consumption. [1]

3.5 Noise analysis

The noise analysis that will follow is based on the assumption that the noise generated in the circuit is white, and the noise sources are independent of each other. As a result, the noise generated by the CSA can be modelled as two noise sources, which contribute in series and in parallel (i_n, u_n).

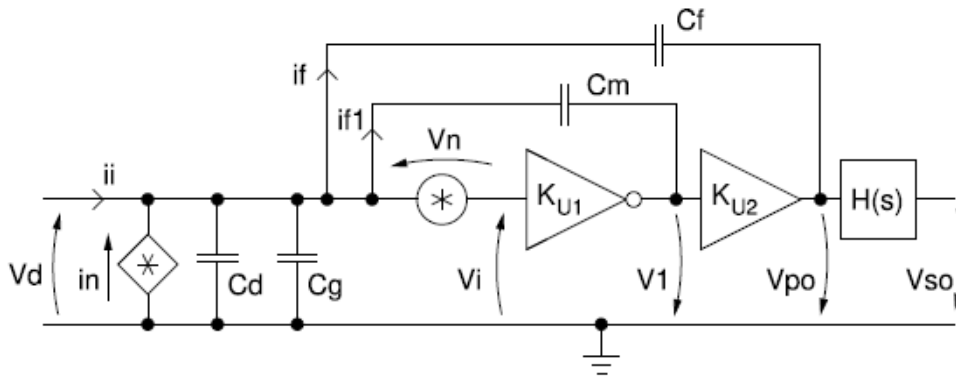


Figure 3.6: CSA schematic for noise analysis. Figure from source [1].

By analysing the circuit in Figure (3.6), the noise can be calculated. It is assumed that the preamplifier has a wide frequency range and that this range is exclusively limited by the shaper. The shaper is a bandpass filter responsible for amplifying the pulse amplitude at the output of the preamplifier and shaping the signal appropriately for the subsequent processing stages.

The parallel and series contributions of the noise are modelled by the current source (i_n) and the voltage source (u_n), respectively. Following an analysis described in Appendix A, the total equivalent noise charge (ENC) is derived. The ENC is the ratio of the rms noise at the output of the shaper over the peak of the circuit's output signal. In order to calculate the series contribution of the noise at the output of the preamplifier, the input current and the parallel noise contribution must be set to zero.

$$ENC_S = \frac{\sigma_{noS}}{u_{soMAX}} = \sqrt{\frac{(2n-2)!}{2n\tau_s} \frac{ne^n C_{in}}{(2n)^n} \overline{u_n}} \quad (3.25)$$

Similarly, by setting the source current (i_i) and the series noise contribution (u_n) to zero, the parallel contribution can be found.

$$ENC_P = \frac{\sigma_{noP}}{u_{soMAX}} = \sqrt{\frac{(2n-1)!n\tau_s}{2}} \left(\frac{e}{2n}\right)^n \overline{i_n} \quad (3.26)$$

The total ENC results from the sum of the parallel ENC and series ENC.

$$ENC^2 = ENC_P^2 + ENC_S^2 \quad (3.27)$$

The detailed calculation of the equivalent noise charge and the concepts of the parameters can be found in Appendix A. [1]

3.5.1 Noise sources in the CSA circuit

The main source of noise, as mentioned, is the input transistor and is responsible for the series noise contribution. It is common for the input transistor's noise to be modelled as a voltage source at the gate of the transistor, and its power spectral density is:

$$S_{in} = \frac{4kT\gamma n}{g_m} \quad (3.28)$$

for which γ is a parameter that takes values from 1/2 to 2/3. Solving for $\bar{u}_i$ and substituting into equation (3.25), we obtain:

$$ENC_S = \sqrt{\frac{(2n-2)!}{2n\tau_s} \frac{ne^n C_{in}}{(2n)^n}} \sqrt{\frac{4kT\gamma n}{g_m}} \quad (3.29)$$

The feedback loop and the sensor's leakage current both contribute in parallel to the noise with spectral power densities:

$$S_{reset} = \frac{4kT}{R_f} \quad (3.30)$$

$$S_{i_{leak}} = 2qI_{leak} \quad (3.31)$$

Solving for $\bar{t}_i$ and substituting into (3.26), we obtain:

$$ENC_{P,reset} = \sqrt{\frac{(2n-1)!n\tau_s}{2}} \left(\frac{e}{2n}\right)^n \sqrt{\frac{4kT}{R_f}} \quad (3.32)$$

$$ENC_{P,leak} = \sqrt{\frac{(2n-1)!n\tau_s}{2}} \left(\frac{e}{2n}\right)^n \sqrt{2qI_{leak}} \quad (3.33)$$

It is worth mentioning that the shot noise of the input MOSFET is considered negligible in radiation detection applications. [1]

Chapter 4

Analysis and simulation of the Charge Sensitive Preamplifier

In the fourth chapter, a schematic level analysis of the CSA is presented. The circuit was designed using XFAB technology at 0.18 μm , and the amplification stage was implemented using a Folded Cascode amplifier with a cascode current source that was analysed extensively in the third Chapter. Additionally, simulations are conducted regarding the performance of the CSA and the contribution of noise to the circuit. Finally, the effect of the detector capacitor and the dimensions of the input and feedback transistors on the circuit's operation is examined.

4.1 Design criteria of the input transistor

Since input transistor plays a crucial role in the contribution of noise, particular emphasis should be given on the dimensions and characteristics of it. To limit the noise, the transconductance of the input transistor should be high, and the input capacitance should be low relative to equation (3.29). As a consequence, special attention should be given to the following. [23]:

- The transistor should be biased with the maximum possible current I_{DS} in order to achieve the highest possible transconductance, with respect to the consumption specifications.
- The transistor should have the smallest possible length in order to minimize

the resistance it exhibits and consequently increase its transconductance.

- Regarding the width of the channel, there are two conflicting principles:
 - The transistor should have a large channel width in order to increase its transconductance.
 - The restriction upon which the input capacity has to be as small as possible set an upper limit to the increase in the channel width.

4.2 Schematic Design of the CSA

The schematic design of the CSA is depicted in figure (4.1), where the transistors' dimensions have been modified in comparison to the design in literature [4] in order to accomplish the preferable outcome.

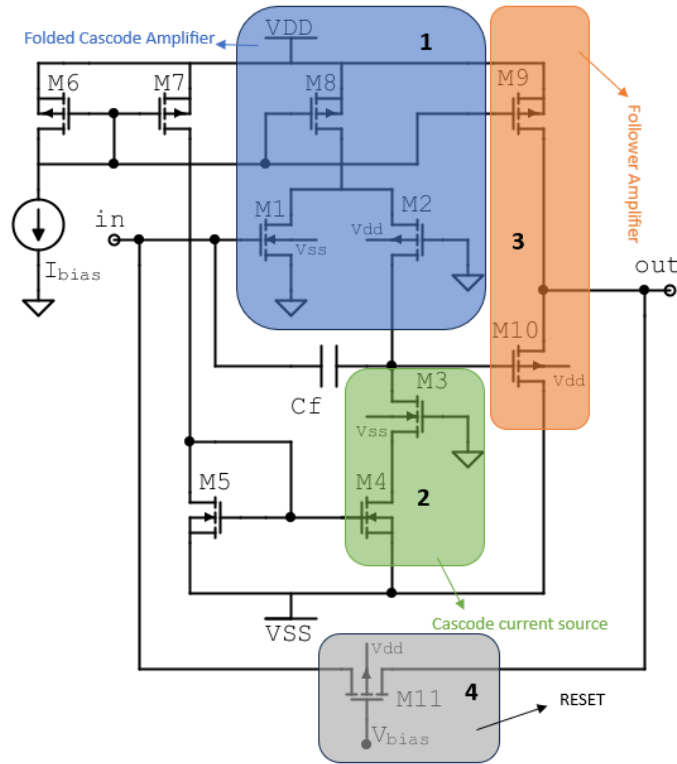


Figure 4.1: CSA schematic with Folded Cascode amplification stage.

4.2.1 Folded Cascode amplifier

In the first block of figure (4.1), the Folded Cascode amplifier is presented. The components of this configuration are as follows:

- Transistor M1: It is the input transistor, which receives the detector's signal at its gate.
- Transistor M2: It is the cascode transistor. It is the common-gate transistor connected to the drain of the common-source transistor (transistor M1).
- Transistor M8: It is the current source that biases the input transistor. Transistor M8 functions as a current mirror with transistor M6, which is biased by the external current source. The relationship of the currents flowing through the transistors in a current mirror is (Appendix C):

$$\frac{I_o}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1} \quad (4.1)$$

All the transistors of this topology operate in the saturation region, with the source-drain current (for the PMOS transistor) or drain-source current (for the NMOS transistor) given by the equations:

$i_D = \frac{1}{2}\mu_n C_{ox}(\frac{W}{L})(u_{GS} - V_{tn})^2, u_{DS} \geq u_{GS} - V_{tn}$	for NMOS transistors
$i_D = \frac{1}{2}\mu_p C_{ox}(\frac{W}{L})(u_{SG} - V_{tp})^2, u_{SD} \geq u_{GS} - V_{tp} $	for PMOS transistors

(4.2)

For which the following apply:

- μ_n and μ_p : They are the mobility of electrons and holes, respectively.
- C_{ox} : It is the oxide capacitance.
- W, L : It is the length and the width of the transistor's channel.
- V_{tn} and V_{tp} : They are the threshold voltages for the NMOS and PMOS transistors, respectively.

It should be mentioned that in order for this circuit to operate, transistor M2 needs to be biased by a current source. This is described in the following paragraph.

The gain achieved and the output resistance for this topology, considering ideal current sources, is (Appendix B): [15]

$$A_u = -g_{m1}g_{m2}r_{ds2}r_{ds1} \quad R_o \simeq g_{m2}r_{ds2}r_{ds1} \quad (4.3)$$

4.2.2 Cascode current source

In the second block of figure (4.1), the cascode current source is depicted. Its main purpose is to bias the transistor M2 so that the Folded Cascode amplifier can operate. Furthermore, using this topology increases the load of the current source, and as a consequence, the amplifier's gain. It is essential to note that in order for the cascode current source to operate, two current mirrors were used: a PMOS current mirror with transistors M6 and M7, and an NMOS current mirror with transistors M5 and M4.

The output resistance and the gain of the Folded Cascode using real current sources are (Appendix B).[15]:

$$A_u = -g_{m1}[g_{m2}r_{ds2}(r_{ds1}||r_{L1})][g_{m3}r_{ds3}r_{ds4}] \quad R_o \simeq [g_{m2}r_{ds2}(r_{ds1}||r_{L1})][g_{m3}r_{ds3}r_{ds4}] \quad (4.4)$$

4.2.3 Voltage Buffer

In the third block of figure (4.1), the voltage buffer is illustrated, which takes at its input the output signal of the Folded Cascode amplifier. The reason why this topology is used is due to its low output resistance and the fact that the voltage at its output is equal to the voltage at its input. This allows large loads to be connected on the preamplifier's output without significantly reducing its gain. The role of each transistor is:

- Transistor M10: It serves as the amplification stage of the voltage buffer. The gain of this amplifier tends toward unity.
- Transistor M9: The role of this transistor is to bias transistor M10. Transistor M9 acts as a current mirror with transistor M6.

In figure (4.2) the small signal analysis of the voltage buffer implemented with PMOS devices is depicted. By examining the schematic in figure (4.2) we will find the gain and the output resistance of the amplifier. Initially, due to the potential difference

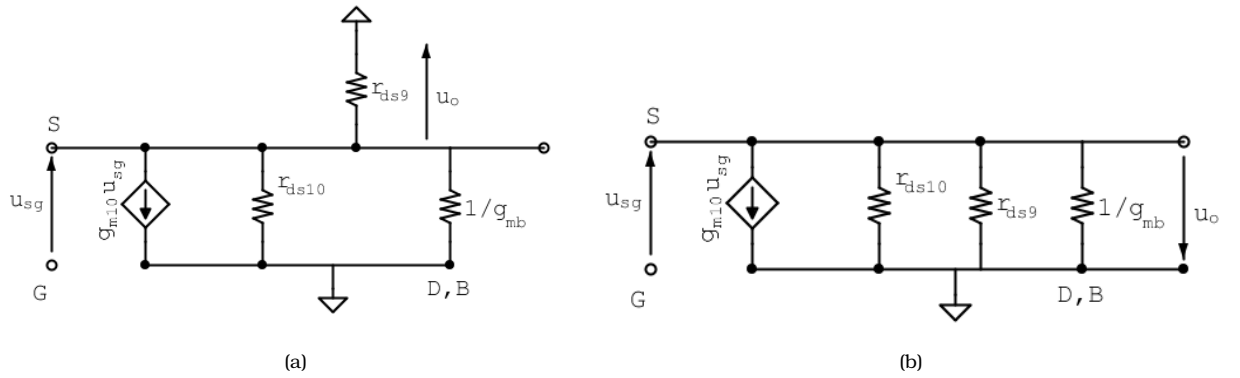


Figure 4.2: (a) Small signal analysis of the voltage buffer. (b) Simplified small signal model of voltage buffer. Both schematics were implemented, considering PMOS transistors for the voltage buffer.

between the substrate and the source of transistor M10, a current is generated. This current should be modelled in the small signal model as a current source. However, this controlled current source appears between the nodes that control it. Therefore, it can be replaced by a resistor with a value of $1/g_{mb}$.

From the analysis of the small signal model, the gain is:

$$\frac{u_o}{u_i} = \frac{u_o}{u_o + u_{SG}} = \frac{g_{m10} u_{SG} R_o}{g_{m10} u_{SG} R_o + u_{SG}} \Rightarrow \frac{u_o}{u_i} = \frac{R_o}{R_o + \frac{1}{g_{m10}}}$$

For which R_o is equal to the parallel combination of the resistances r_{ds10} , r_{ds9} , $1/g_{mb}$. However, usually the resistance $1/g_{mb}$ is much smaller than the others, thus:

$$R_o \simeq 1/g_{mb} \quad \frac{u_o}{u_i} \simeq g_{m10}/(g_{mb} + g_{m10}) \quad (4.5)$$

It is important to stress that the transconductance due to the substrate and the

transconductance due to the gate are related by the equation $g_{mb} = xg_{m10}$, where x takes values from 0.1 to 0.2. We can therefore conclude that the gain converges quite well to unity.[15]

4.2.4 Feedback

The feedback resistance of the CSA is modelled by a PMOS transistor, which operates in the linear region. The reason of using a resistive transistor rather than a resistor is that the resistor occupies significant space in the physical layout and introduces more thermal noise into the circuit. The current in the linear region of a PMOS transistor is:

$$i_D = \mu_p C_{ox} \left(\frac{W}{L} \right) [(u_{SG} - |V_{tp}|)u_{SD} - (1/2)u_{SD}^2] \quad , u_{SD} < |u_{GS} - V_{tp}| \quad (4.6)$$

4.3 Dimensions and Specifications

The specifications of the detector are shown in Table (4.1), and the time between two events is at least 1 ms. Both the specifications in Table (4.1) and the time interval between two neighbouring events are consistent with the literature ([23]).

Table 4.1: Detector's specifications

Detector	
Detector's capacitance	$C_d = 2 \text{ pF}$
Leakage current	$I_{leak} = 10 \text{ pA}$
Collected charge per event	$Q_{typical} = 42000e^-$ $Q_{max} = 310000e^-$
Time needed for the collection of 90% of the charge	300 ns

Furthermore, the transistors' dimensions are depicted in table (4.2).

Table 4.2: Dimensions of NMOS and PMOS transistors.

transistor	W/L $\mu\text{m}/\mu\text{m}$
M1	310/0.9
M2	100/0.7
M3	25/1.5
M4	2.5/5
M5	2.5/5
M6	2.5/2.5
M7	2.5/2.5
M8	5/2.5
M9	2.5/2.5
M10	100/0.7
M11	1/6

4.4 Simulation of the Schematic

The circuit that was simulated is the one shown in Figure (4.1). The supply voltages, the bias current I_{BIAS} , the bias potential of the feedback transistor M11, the value of the feedback capacitor, and the detector capacitor are presented in Table (4.3). Additionally, by substituting the charge collected in each event into the current formula (4.7), the detector's current pulse is obtained.

Table 4.3: Power supply and biasing.

Positive potential	$V_{DD} = 1.65 \text{ V}$
Negative potential	$V_{SS} = -1.65 \text{ V}$
Biasing current	$I_{BIAS} = 10 \text{ uA}$
Biasing voltage	$V_{BIAS} = 300 \text{ mV}$
Feedback capacitor	$C_f = 200 \text{ fF}$
Detector's capacitance	$C_d = 2 \text{ pF}$

In this implementation, 90% of the total charge ($310000e^-$) is collected in 300 ns. Therefore:

$$I = \frac{dQ}{dt} = \frac{0.9 * 310000e^-}{3 * 10^{-7}} = 148.8nA \quad (4.7)$$

4.4.1 Time response

As described in the first chapter, the output of the preamplifier is expected to be a voltage pulse. More specifically, as long as there is no detector current at the

input, the circuit remains in balance, with the output voltage and the input voltage being equal. However, when input current appears, a small incremental voltage is generated at the gate of the input transistor. This voltage signal is significantly amplified by the Folded Cascode, resulting in a lower potential at the output.

The conditions are now met for the capacitor to be charged by the current from the detector. When the current ceases to exist, the charging of the capacitor stops, and the circuit attempts to return to equilibrium by discharging the capacitor from the input to the output through the resistive transistor M11. It has to be noted that the current pulse shows a delay of 10 μs for waveform display purposes.

Furthermore, through the simulation environment, it was calculated that the charging time of the capacitor is 232.1 ns. This confirms that a charge collected in 300 ns can indeed be detected. The discharge time was calculated to be 933.6 ns, which, although long, does not pose a problem because the specifications of this particular application indicate that the time interval between two consecutive events is at least 1 ms. The output waveform and input current diagram is presented in Figure (4.3). In this, the inversion of the output signal becomes apparent, since the output signal has a phase difference of 180° from the input signal.

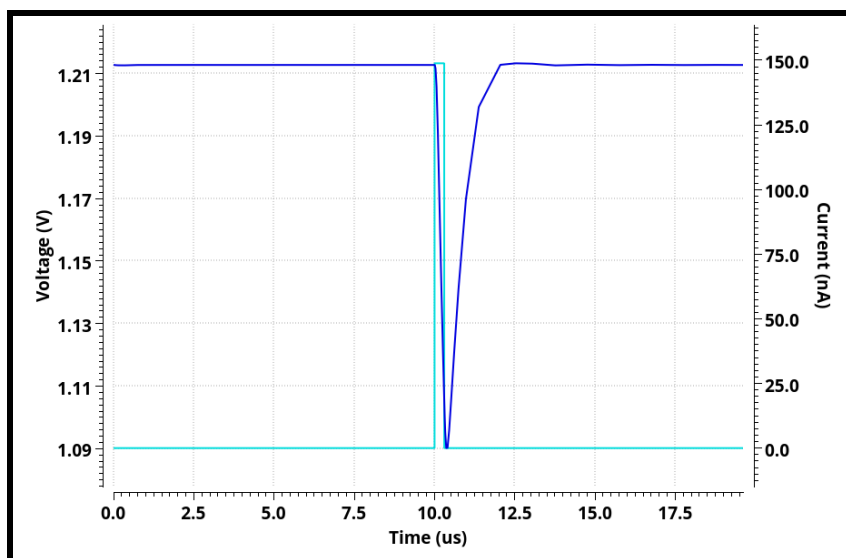


Figure 4.3: Time response of the CSA's schematic.

4.4.2 Frequency response

In Figure (4.4), the Bode diagram of the CSA is depicted. By conducting measurements in the simulation environment, the gain of the preamplifier and its operating range were determined.

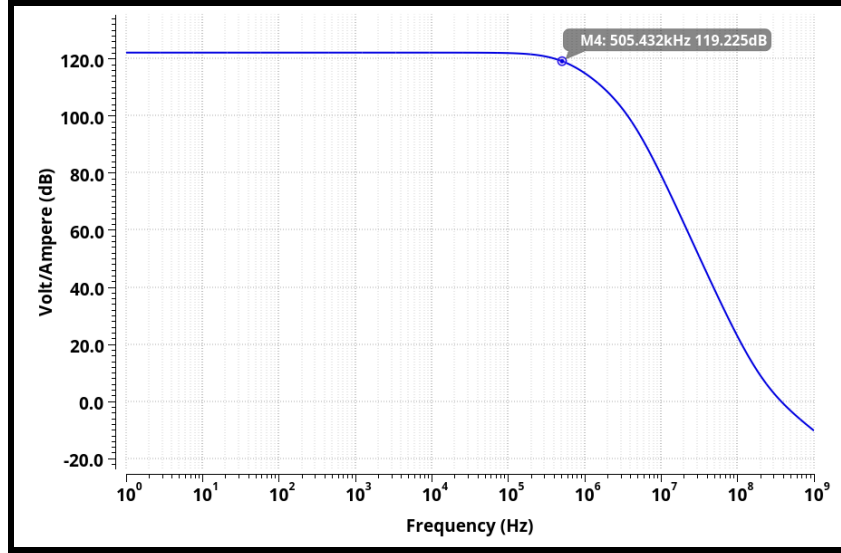


Figure 4.4: Frequency response of the CSA's schematic.

$$A = 122.23 \text{ dB} \quad BW = 505.43 \text{ kHz} \quad (4.8)$$

4.4.3 Noise Analysis

It is undoubtedly known that white noise and flicker noise are observed everywhere in CMOS technology, and as a result, a diagram for the noise Power Spectral Density [PSD] is expected to emerge similar to the one in paragraph 2.2 (2.2). In figure (4.5), the diagram of the noise PSD as a function of frequency is depicted. Specifically, the presence of flicker noise at frequencies lower than the corner frequency (around 1 kHz) becomes apparent, as the PSD is inversely proportional to the frequency in that region. Furthermore, at frequencies greater than the corner frequency and less than the cut-off frequency (505.43 kHz), thermal noise dominates, as the PSD remains constant. Finally, for frequencies higher than the cut-off frequency, the noise decreases as we move away from the circuit's operating range.

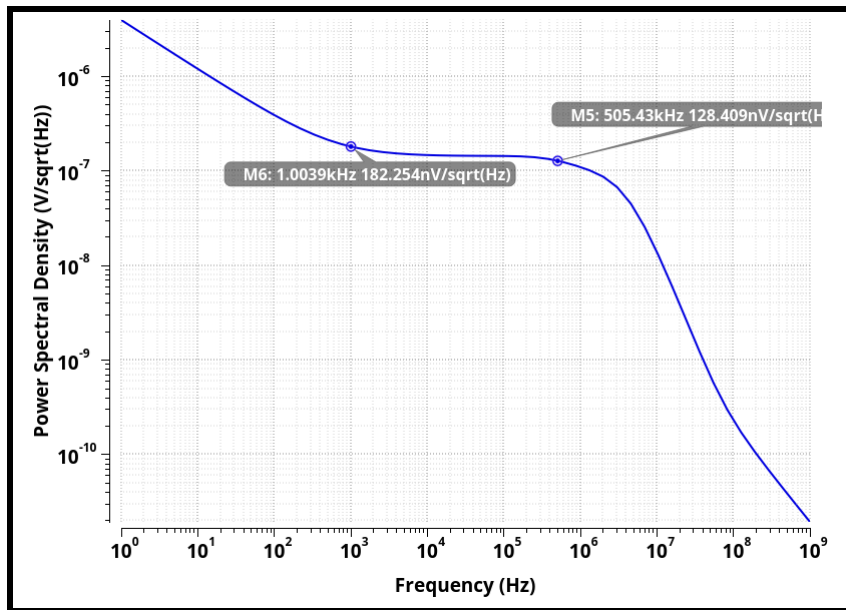


Figure 4.5: PSD diagram of noise voltage.

4.4.4 Linearity

By gradually increasing the collected charge and measuring the peak of the output waveform, the linearity of the circuit can be determined. More specifically, by applying the method of least squares in the collected data (figure 4.6), the growth of the voltage as a function of the charge was found to be 2.29 mV/fC.

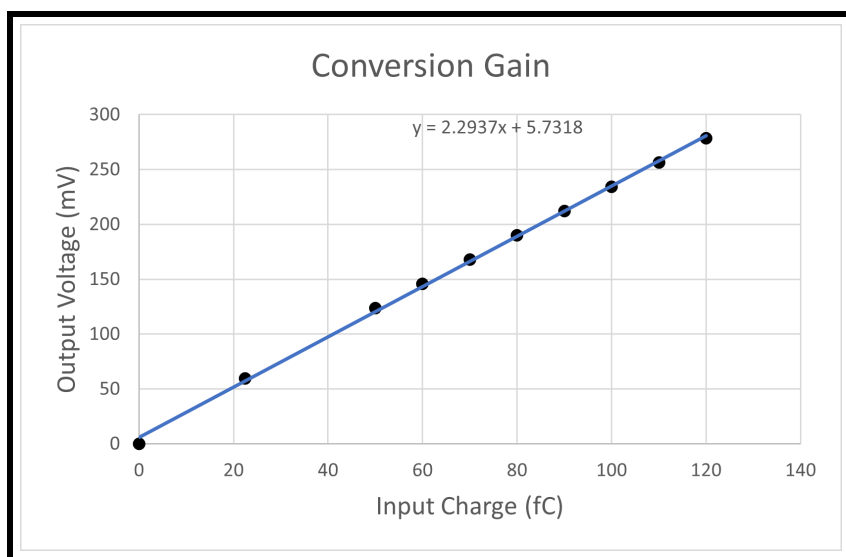


Figure 4.6: Diagram of peak output voltage as a function of the charge.

4.5 Parametric Analysis

4.5.1 Varying detector's capacitance

Since the detector capacitance is modeled as a capacitor in parallel to a current source, the effect in the output signal when the capacitor value is varying will be examined. The capacitor values for all subsequent parametric analyses are: 2 pF, 18 pF, 34 pF, and 50 pF.

4.5.1.1 Time response

The time response of the parametric capacitance analysis of CSA is depicted in figure (4.7). In the diagram, it is observed that as the detector's capacity increases,

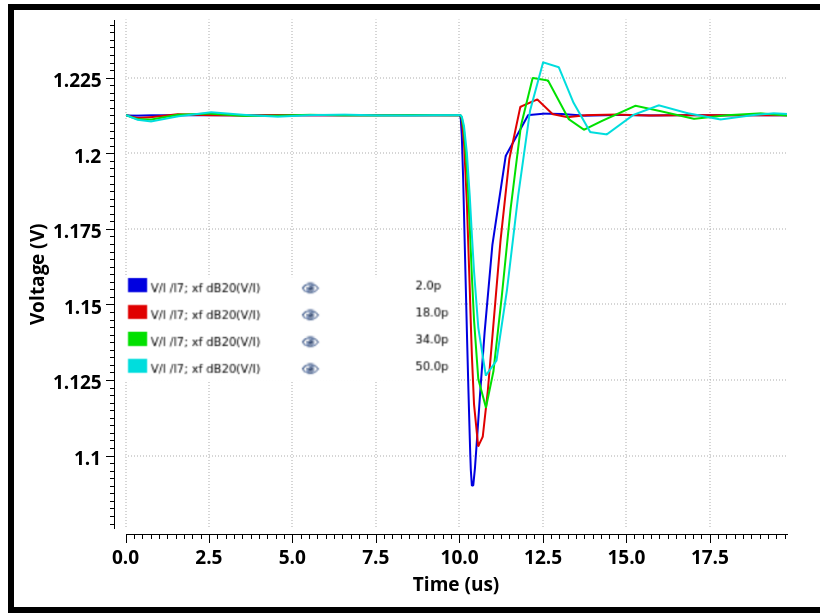


Figure 4.7: Time response of the CSA for varying detector's capacitance values.

the output signal becomes distorted. This is an indication that the circuit is no longer stable.

4.5.1.2 Frequency response

In figure (4.8), the frequency response of the parametric capacitance analysis of CSA is depicted. In the Bode diagram, a peak is observed near the cut-off frequency. This is an indication that the circuit is becoming unstable as the detector's

capacitance increases. More information on why this occurs can be found in source [24], specifically in the section concerning the complex poles of the transfer function (when the denominator of the transfer function is of second order or higher, as found in Chapter 1 1.11, it may exhibit complex poles).

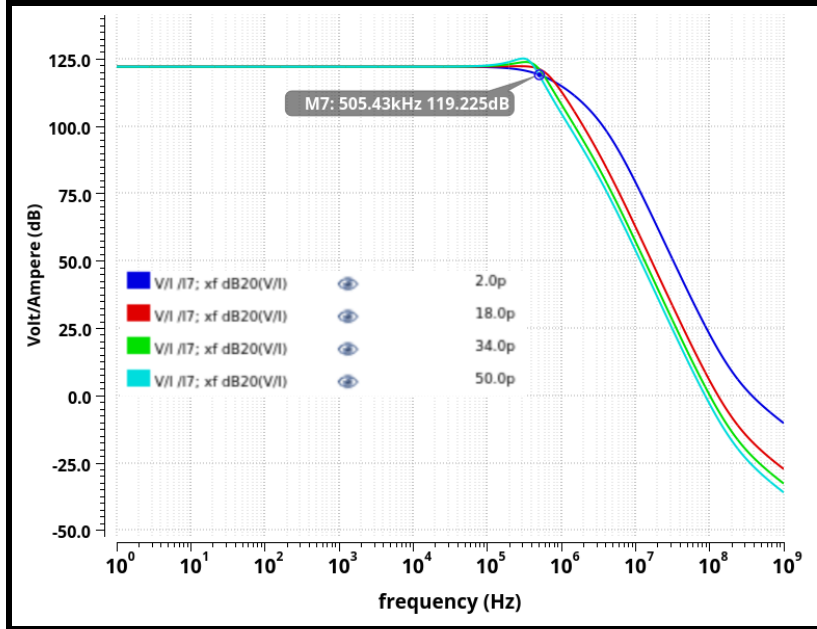


Figure 4.8: Frequency response of the CSA for varying detector's capacitance values.

4.5.1.3 Noise analysis

In figure (4.9), the parametric noise analysis of the circuit is depicted. The PSD diagram of the noise illustrates similar behavior to that of the frequency response. Specifically, in the cut-off frequency range, a peak in the PSD is observed as the capacitance of the detector increases.

4.5.2 Varying the input transistor's surface area

In this paragraph, the effect of changing the dimensions of the input transistor is examined. In particular, the length and width of the channel are increased proportionally, keeping the ratio of the two dimensions constant. Table (4.4) shows the dimensions of the input transistor, where it becomes evident that the coefficient of the initial length and the initial width increases by 0.5 in each simulation. The di-

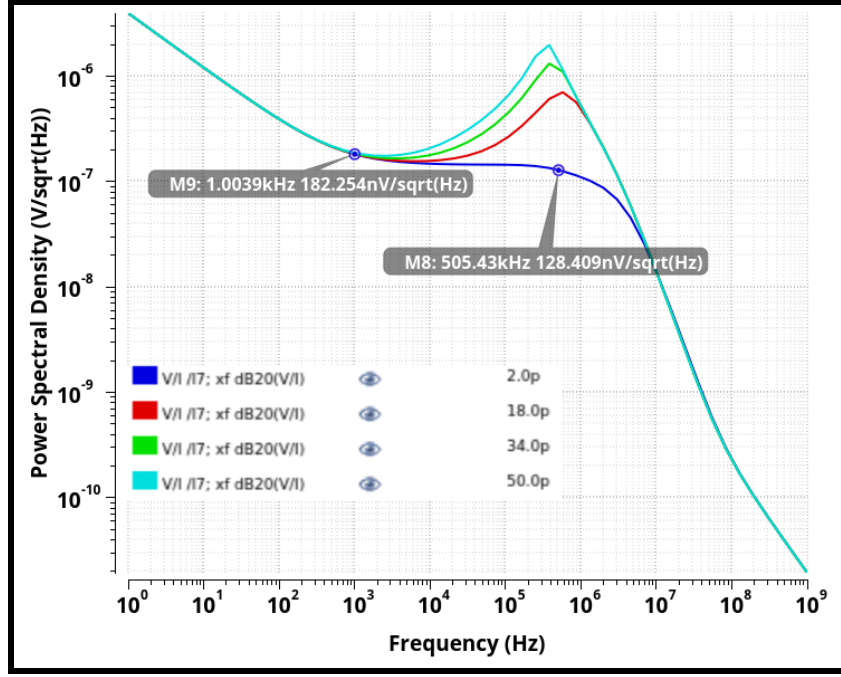


Figure 4.9: Noise Power Spectral Density of the CSA for varying detector's capacitance values.

mensions of the cascode transistor (M2) should be changed by the same coefficient.

Table 4.4: Input transistor's dimensions.

Input transistor (M1)			Cascode transistor (M2)	
Coefficient (a) of W & L	W_1	L_1	W_2	L_2
1	310 μm	0.9 μm	100 μm	0.7 μm
1.5	465 μm	1.35 μm	150 μm	1.05 μm
2	620 μm	1.8 μm	200 μm	1.4 μm
2.5	775 μm	2.25 μm	250 μm	1.75 μm

4.5.2.1 Time response

In the diagram (4.10), the time response of the circuit when the dimensions of the input transistor are increasing while keeping their ratio constant is depicted. It is worth mentioning that the transconductance of the input transistor remains constant, as the ratio of the dimensions does not change. The diagram shows a very slight increase in the DC level, which does not pose a problem for the circuit's operation.

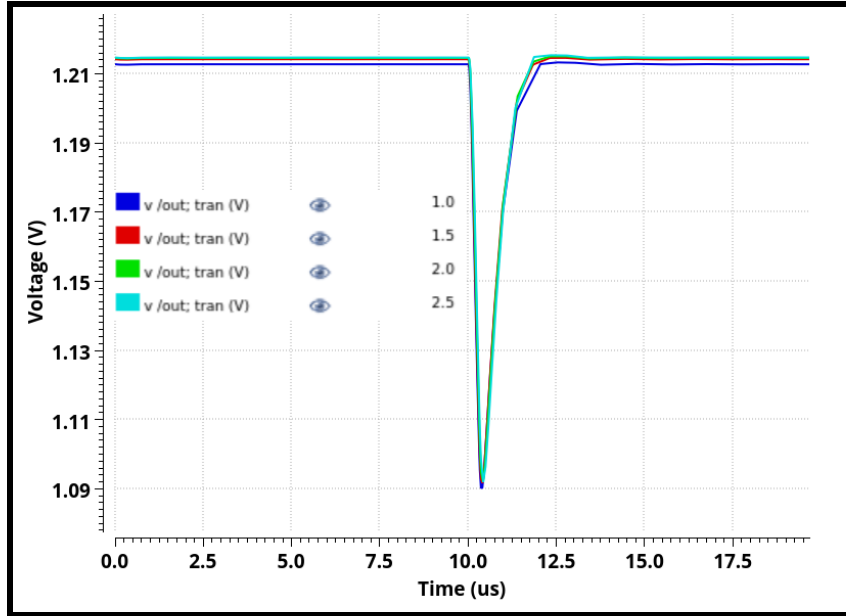


Figure 4.10: Time response of the CSA for varying the input transistor's area.

4.5.2.2 Frequency response

In the diagram (4.11), the frequency response of the circuit is illustrated for varying surface area of the input transistor. It is observed that the gain decreases at a slightly faster rate as the channel length increases. This is likely due to the shifting of higher frequency poles to lower frequencies. The gate-source and gate-drain capacitances of the input transistor increase, resulting in an increase in the input capacitance. Consequently, the second pole of the amplification stage of the circuit shifts to lower frequencies, as indicated by the equation (3.8).

4.5.2.3 Noise analysis

In Figure (4.12), the PSD diagram of the parametric analysis of the input transistor's surface area is depicted. It is observed that as the area of the input transistor's channel increases, the flicker noise decreases and the white noise increases. The reduction in flicker noise can be explained by equation (2.14), which describes the root mean square value of the flicker noise voltage. This equation shows an inverse relationship between the root mean square value of the noise voltage and the dimensions of the transistor; thus, increasing the dimensions leads to a reduction in flicker noise. The reduction in flicker noise results in an increase of the white noise

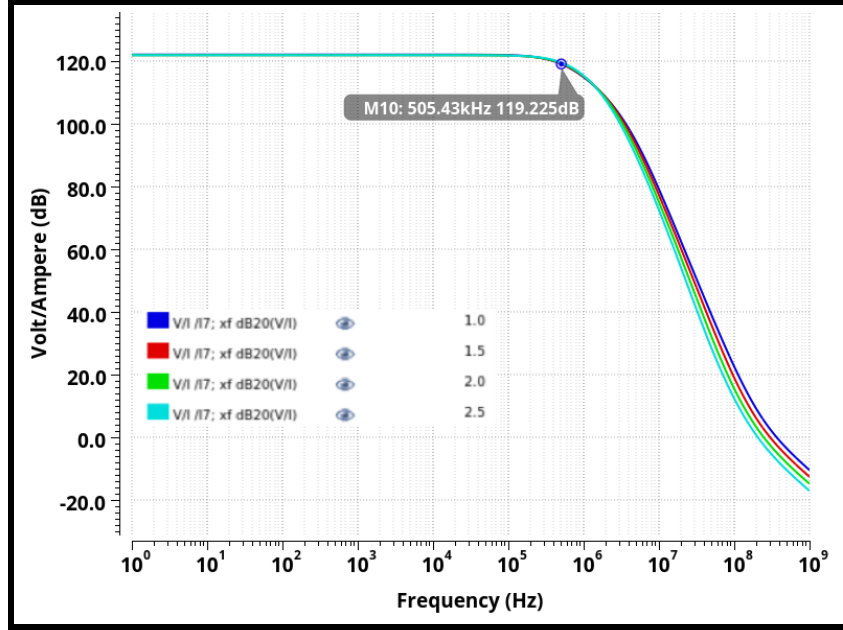


Figure 4.11: Frequency response of the CSA for varying the input transistor's area.

contribution in the circuit.

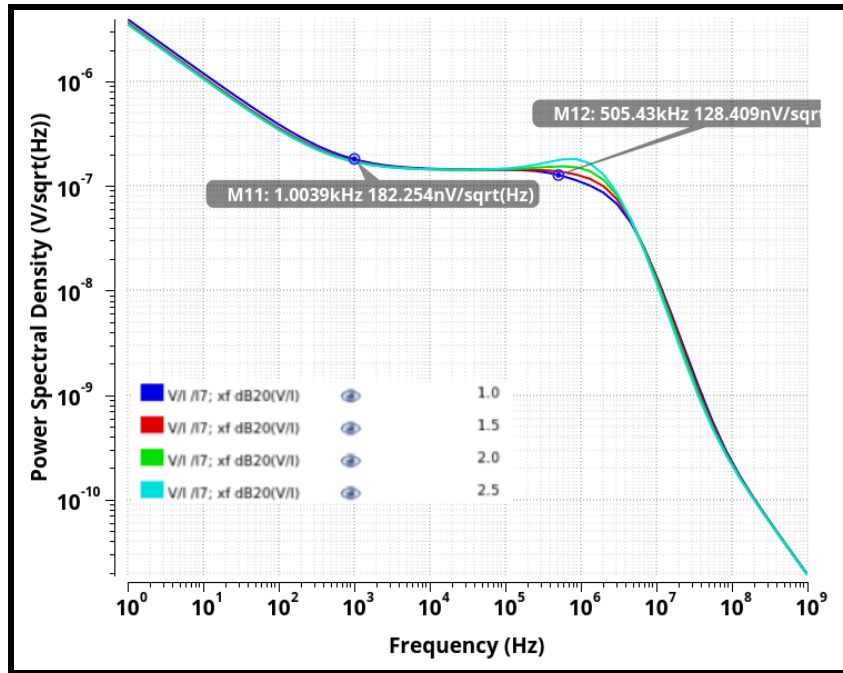


Figure 4.12: Noise Power Spectral Density of the CSA for varying the input transistor's area.

4.5.3 Varying the feedback transistor's surface area

In this paragraph, the dimensions of the feedback transistor are increased while keeping their ratio constant. The dimensions of the feedback transistor are shown

in Table (4.5), where it can be observed that the coefficient of the initial length and the initial width increases by 0.5 in each simulation.

Table 4.5: Feedback transistor's dimensions.

Feedback transistor (M11)		
Coefficient (a) of W and L	W	L
1	1 μm	6 μm
1.5	1.5 μm	9 μm
2	2 μm	12 μm
2.5	2.5 μm	15 μm

4.5.3.1 Time response

In the figure (4.13), the time response of the CSA for varying the feedback transistor's surface area is depicted. It is observed that the output waveform of the circuit is not affected by this parametric change. Similar to the previous analysis, the transconductance of the feedback transistor remains constant because the ratio of the transistor's dimensions is unchanged. As a result, the feedback transistor's resistance remains steady.

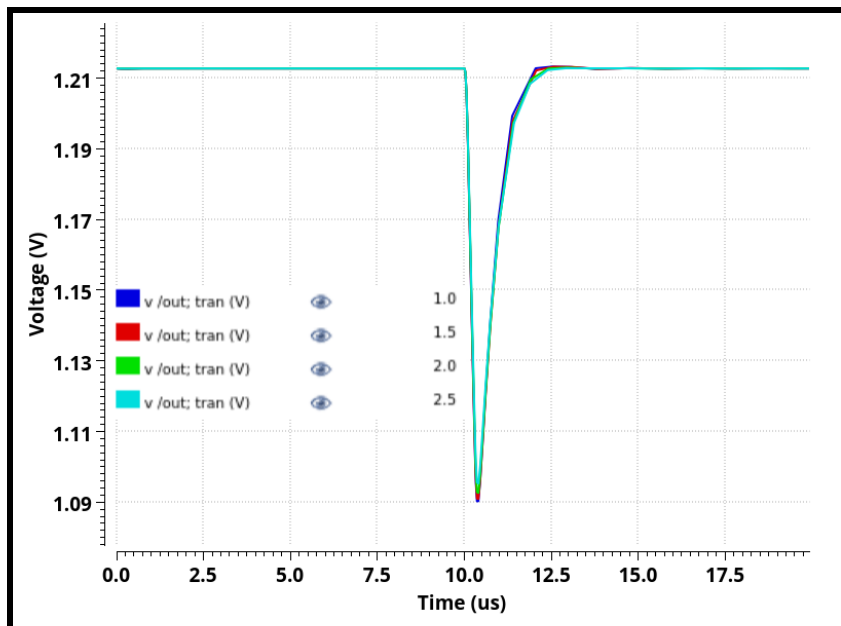


Figure 4.13: Time response of the CSA for varying the feedback transistor's area.

4.5.3.2 Frequency response

In Figure (4.14), the frequency response of the CSA for varying the feedback transistor's surface area is depicted. It becomes clear from the diagram that the change in the dimensions of the feedback transistor, while keeping their ratio constant, does not have a significant effect on the frequency response.

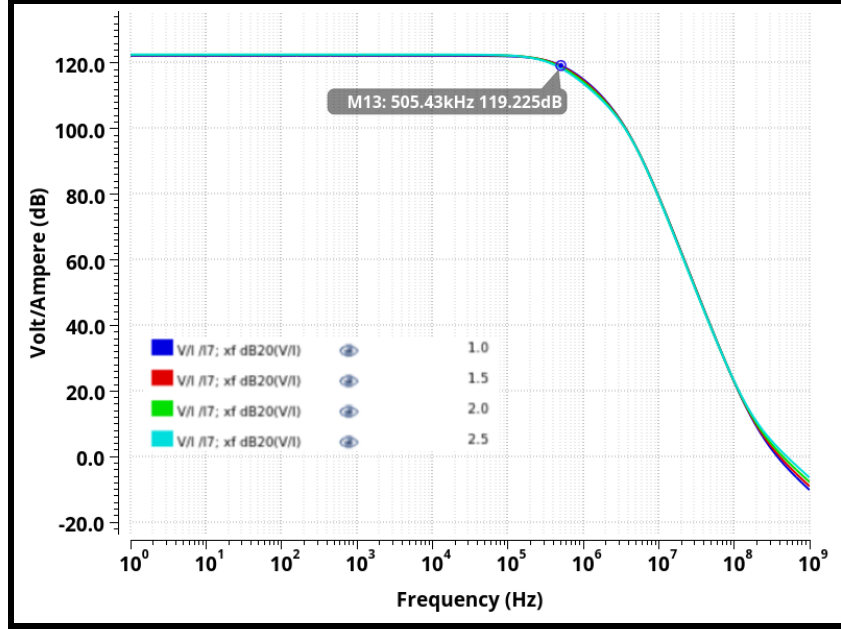


Figure 4.14: Frequency response of the CSA for varying the feedback transistor's area.

4.5.3.3 Noise analysis

In Figure (4.15), the noise Power Spectral Density of the CSA for varying the feedback surface transistor's area is depicted. Observing the diagram, it appears that increasing the surface area of the feedback transistor does not affect the noise contribution in the circuit. As mentioned in Chapter 3, the resistance of the feedback (transistor M11) introduces thermal noise into the circuit. However, since the resistance of the feedback transistor remains constant, because the dimensions' ration of the transistor is unchanged, the noise contribution of the transistor M11 remains the same.

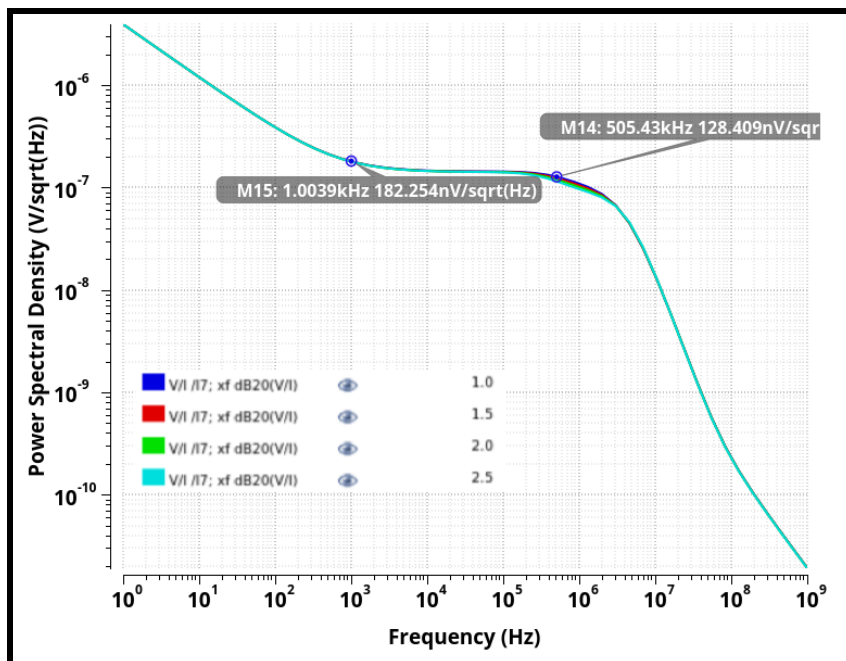


Figure 4.15: Noise Power Spectral Density of the CSA for varying the feedback transistor's area.

Chapter 5

The impact of Layout design in the performance of the CSA

In the fifth chapter, the layout design of the CSA is presented (figure 5.1). The implementation of the layout design was carried out using the 0.18 μm technology from XFAB in the Cadence's Virtuoso environment.

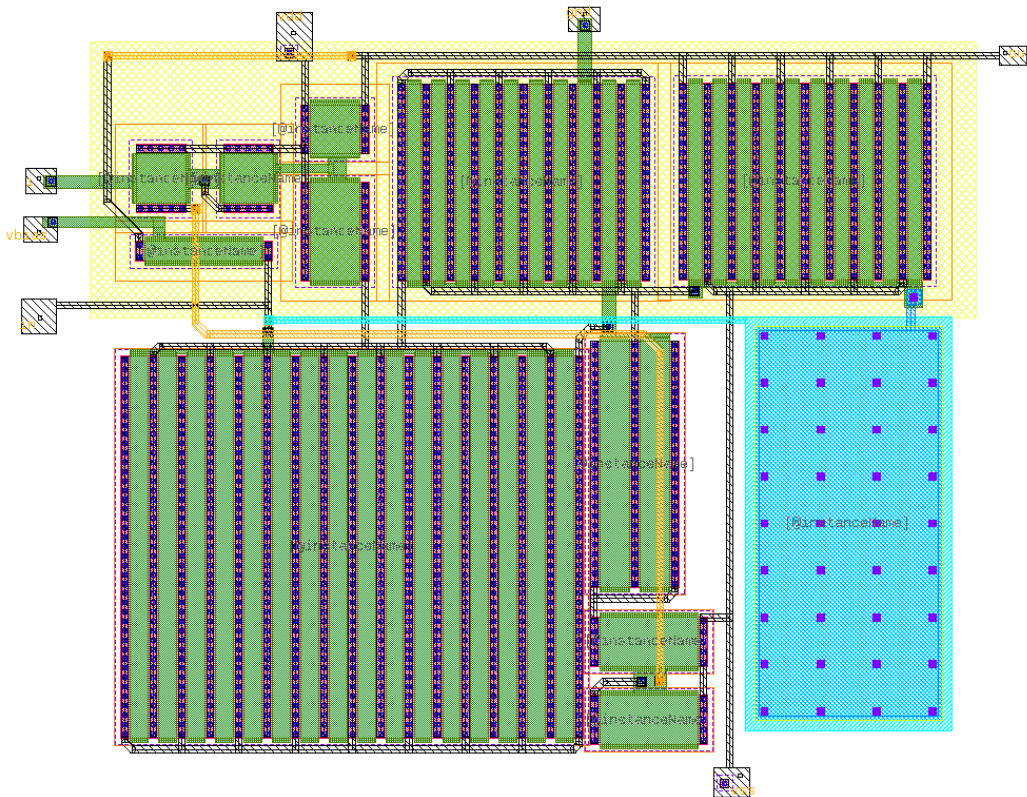


Figure 5.1: Layout design of the CSA.

5.1 Simulation of the CSA including the impedances of the layout design's interconnections

In the following paragraphs, the performance of the CSA is examined after the implementation of the layout design (Post Layout Simulation) to include the physical interconnections' impedances. Similarly to the fourth chapter, the time response, frequency response, and the noise Power Spectral Density diagram are extracted. The results of the simulations are discussed and compared with those from the previous chapter.

5.1.1 Time response

In the diagram (5.2), the time response of the CSA is depicted, both before and after the layout design. After the layout design, it is apparent that a small increase has formed in the DC level of the output waveform by 2.7 mV. In addition, a slight time delay was noted in the sharp decrease of the output signal just after 10 us. Furthermore, it is evident that the feedback capacitor discharges faster after the layout design, since the output signal rise back to approximately 1.21 V in a shorter period of time.

5.1.2 Frequency response

In Figure (5.3), the frequency response of the CSA is depicted. In this simulation, the waveforms appear quite similar; however, certain variations have to be mentioned. In particular, the frequency range slightly increases after layout design, while the gain decreases in a small extent.

The measured values with the help of the simulation environment are:

$$A = 121.96dB \quad BW = 526.76kHz \quad (5.1)$$

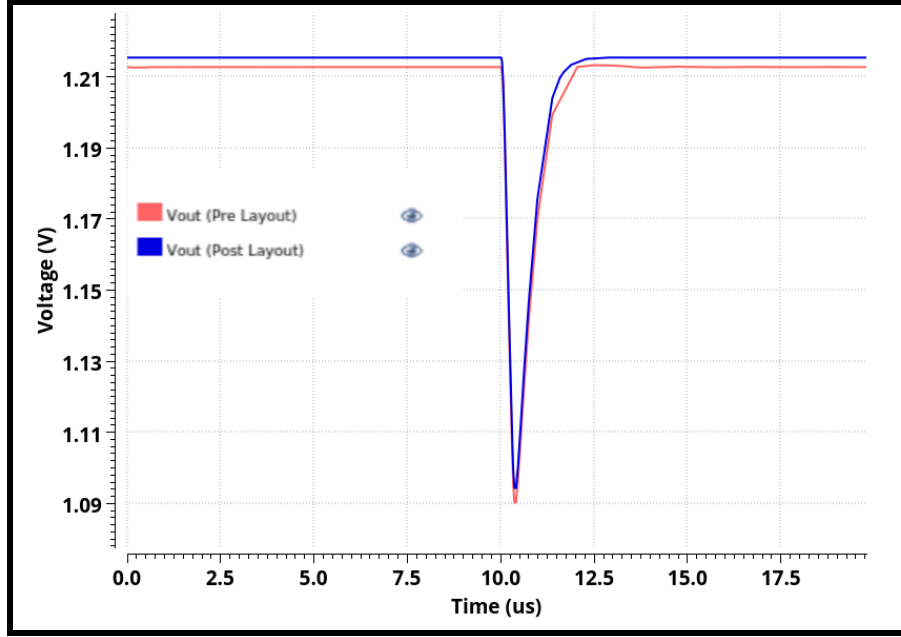


Figure 5.2: Post layout simulation of the CSA's time response.

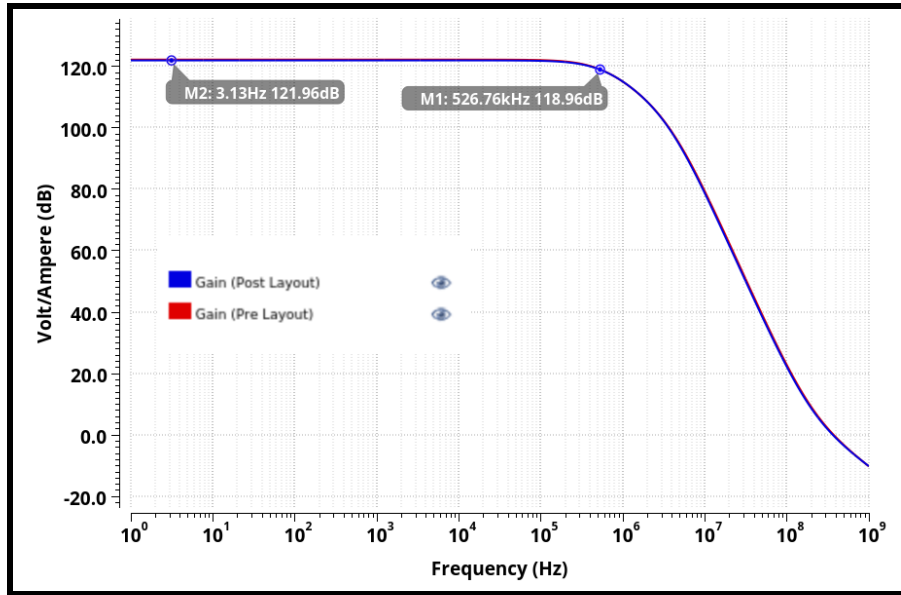


Figure 5.3: Post layout simulation of the CSA's frequency response.

5.1.3 Noise analysis

In the figure (5.4), the PSD diagrams of the noise before and after the layout design are illustrated. If we focus on the two waveforms, it becomes clear that the post layout waveform exhibits greater noise both at low frequencies, below the corner frequency, and at high frequencies, between the corner frequency and the cut-off frequency. Consequently, the values of the PSD at both the cut-off frequency and the corner frequency were higher after the layout design (Table 5.1).

5.1. Simulation of the CSA including the impedances of the layout design's interconnections

Table 5.1: Feedback transistor's dimensions

	Pre layout PSD	Post layout PSD
Corner frequency	182.254 nV/ $\sqrt{\text{Hz}}$	183.747 nV/ $\sqrt{\text{Hz}}$
Cut-off frequency	128.409 nV/ $\sqrt{\text{Hz}}$	132.075 nV/ $\sqrt{\text{Hz}}$

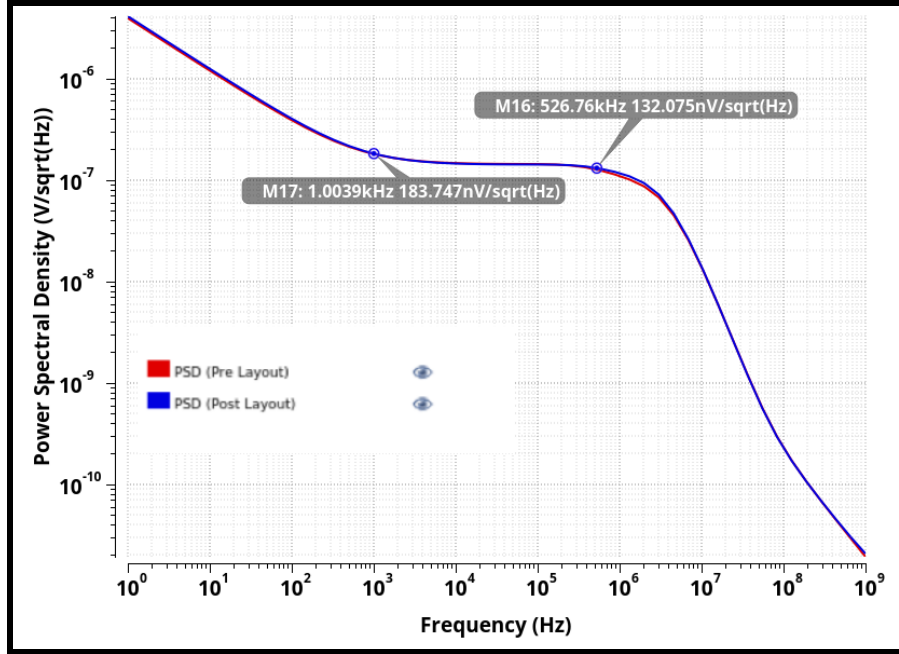


Figure 5.4: Post layout simulation of the CSA's noise Power Spectral Density.

In conclusion, the impedance of the interconnections from the layout design does not significantly affect the performance and operation of the circuit. Therefore, the layout design can be considered sufficient.

Chapter 6

Bachelor thesis's conclusions

In the last chapter, the most important points and conclusions of the CSA circuit for radiation and particle detection are discussed.

Initially, the two most common types of detectors used in High Energy Physics experiments were analysed; these are the strip detectors and pixel detectors. The analysis revealed that pixel detectors have significant advantages over the other type of detectors. A key characteristic of pixel detectors is that they exhibit lower input capacitance and smaller leakage current due to the smaller size of the pn diodes.

Afterwards, the gain stage used in the Charge Sensitive Preamplifier was studied. It was decided that the topology of the Folded Cascode amplifier is the most suitable compared to the Cascade and Cascode topologies. In particular, both the Folded Cascode and Cascode topologies have advantages over the Cascade topology for three main reasons: the Miller effect is limited to a greater extent, they behave similarly to a single-stage amplifier as they have a dominant low-frequency pole, and the noise contribution is confined to the input transistor, thereby improving the signal-to-noise ratio (SNR). Additionally, it was decided to use a cascode current source to bias the cascode transistor. This choice provides greater gain while keeping the gain-bandwidth product stable and does not further contribute to the Miller effect.

In the following chapters of the bachelor thesis, the CSA circuit was simulated and the impact of certain parameters was investigated. Firstly, by increasing the

value of the detector capacitance the output signal was distorted, which became evident from the time response, the frequency response, and the noise PSD diagram respectively. Specifically, for the time response, the output voltage was distorted in the discharge of the feedback capacitor, and in the following two diagrams both the gain and the noise PSD had reached a peak at the cut-off frequency.

Another parameter that was altered was the surface area of the input transistor's channel. In particular, the length and width of the input transistor were increased, keeping the aspect ratio constant. In this case, the circuit's output signal remained unaffected. It was also observed that the gain signal decayed more quickly with the increase in the input transistor channel's surface area. Finally, it was noted that the white noise increased, while the flicker noise decreased. In the case of flicker noise, the increase in the transistor's dimensions led to its reduction, according to equation (2.14), which justifies the result of the diagram. The reduction in flicker noise resulted in an increase in the contribution of white noise.

The last parameter that was modified was the surface area of the feedback transistor's channel. The way in which the surface area of the feedback transistor's channel was varied remained similar to that of the input transistor's channel. In this case, the time and frequency response remained unchanged, as well as the noise Power Spectral Density. It should be stressed that the resistance of the feedback transistor significantly affects the circuit's simulation that were conducted in this thesis. However, as the channel dimensions changed proportionally, the transistor's transconductance remained constant, and as a consequence, the resistance presented by the feedback transistor remained stable. This is the reason that the output waveform, the noise PSD, and the frequency response remained unchanged.

In the last stage of the bachelor thesis, the Charge Sensitive Preamplifier circuit was simulated, including the impedances of the physical interconnections that resulted from the layout design. It was apparent that the time response, frequency response, and noise PSD did not change significantly. This led to the conclusion

that the implemented layout design was sufficient, since it did not have a significant impact on the operation and performance of the circuit.

Appendix A

Calculation of the Equivalent noise charge

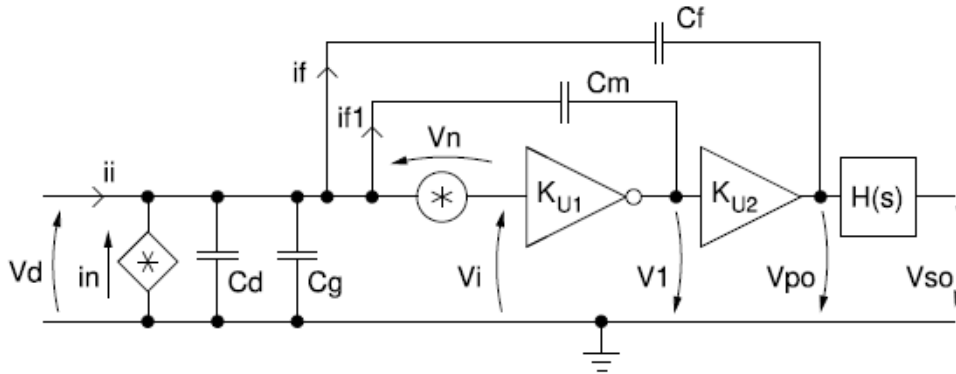


Figure 1: CSA schematic for the noise study. Figure from source [1].

By analyzing the schematic in figure (1), the following equations can be extracted:

$$u_n = u_d - u_i \quad (\text{A.1}) \quad i_f = sC_f(u_d + u_{po}) \quad (\text{A.2})$$

$$i_{f1} = sC_m(u_d + u_1) \quad (\text{A.3}) \quad s(C_d + C_g)u_d + i_f + i_{f1} = i_i + i_n \quad (\text{A.4})$$

$$u_1 = u_i K_{U1} \quad (\text{A.5}) \quad u_{po} = u_1 K_{U2} \quad (\text{A.6})$$

$$K = K_{U1} K_{U2} \quad (\text{A.7})$$

In order to find the output voltage of the preamplifier, the noise contributions need

to be set to zero. Considering a Dirac current pulse at the input of the preamplifier, the output voltage is calculated using the above equations as:

$$\begin{aligned} u_{po} &= i_i \frac{1}{s(C_f + C'_{in}/K_U)} \\ C'_{in} &= C_d + C_g + C_f + (1 + K_{U1})C_m \end{aligned} \quad (\text{A.8})$$

In order to find the output voltage in the time domain, the output voltage of the preamplifier must be multiplied by the transfer function of the shaper and the Laplace transform must be applied. The transfer function of the shaper is:

$$H(s) = \frac{s\tau_s}{(1 + s\tau_s)^{n+1}} \quad (\text{A.9})$$

Therefore, the voltage of the overall circuit is obtained as:

$$u_{so}(t) = \frac{Q_i}{C_f + C'_{in}/K_U} \frac{1}{n!} \left(\frac{t}{\tau_s}\right)^n \exp\left(-\frac{t}{\tau_s}\right) \quad (\text{A.10})$$

This waveform exhibits a peak at a time equal to the peaking time. The peaking time is the time it takes for the output waveform to transition from the resting voltage to its maximum value and is given by the following equation:

$$t_{peak} = n\tau_s \quad (\text{A.11})$$

For which:

- n : It is the order of the shaping filter.
- τ_s : It is the time constant of the differentiator and integrators of the shaper.

The maximum value of the output waveform is:

$$u_{soMAX} = \frac{Q_i}{C_f + C'_{in}/K_U} \frac{1}{n!} \left(\frac{v}{e}\right)^n \quad (\text{A.12})$$

As mentioned in the third Chapter, to determine the series contribution of the noise,

both the input current and the parallel contribution of the noise must be set to zero.

$$u_{po} = u_n \frac{C_{in}}{C_f + C'_{in}/K_U} \quad (\text{A.13})$$

For which it applies:

$$C_{in} = C_d + C_g + C_f + C_m$$

$$C'_{in} = C_d + C_g + C_f + (1 + K_{U1})C_m$$

Transforming the equation (A.13) in the frequency domain, then multiplying by the transfer function of the shaper, and finally integrating over the entire frequency spectrum, yields the root mean square [RMS] value of the in series noise source:

$$\sigma_{noS}^2 = \int_0^\infty \frac{\overline{u_n^2}}{\Delta f} \frac{C_{in}^2 |H(j\omega)|^2}{(C_f + C'_{in}/K_U)^2} df \quad (\text{A.14})$$

Dividing the RMS value ($\sqrt{\sigma_{noS}^2}$) by the maximum output voltage of the shaper yields the total Equivalent Noise Charge [ENC] for the in series noise source:

$$ENC_S = \frac{\sigma_{noS}}{u_{soMAX}} = \sqrt{\frac{(2n-2)!}{2n\tau_s}} \frac{ne^n C_{in}}{(2n)^n} \overline{u_n} \quad (\text{A.15})$$

In order to determine the parallel noise contribution, the input current and the in series noise contribution must be set to zero.

$$u_{po} = i_n \frac{1}{s(C_f + C'_{in}/K_U)} \quad (\text{A.16})$$

By transforming the relationship (A.16) in the frequency domain, then multiplying by the transfer function of the shaper and finally integrating over the entire frequency spectrum, the RMS value of the parallel noise source is obtained:

$$\sigma_{noP}^2 = \int_0^\infty \frac{\overline{i_n^2}}{\Delta f} \frac{C_{in}^2 |H(j\omega)|^2}{(C_f + C'_{in}/K_U)^2 \omega^2} df \quad (\text{A.17})$$

Dividing the RMS value ($\sqrt{\sigma_{noP}^2}$) by the maximum output voltage of the shaper yields

the total ENC for the parallel noise source is calculated.

$$ENC_P = \frac{\sigma_{noP}}{u_{soMAX}} = \sqrt{\frac{(2n-1)!n\tau_s}{2}} \left(\frac{e}{2n}\right)^n \bar{t}_n \quad (\text{A.18})$$

Appendix B

Calculations for the Folded Cascode gain

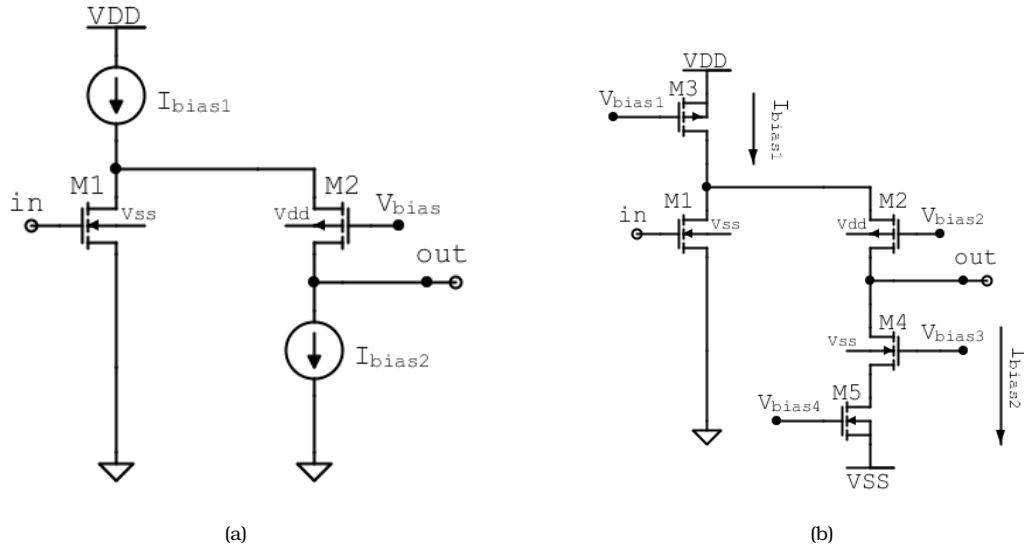


Figure 1: (a) Folded Cascode amplifier, implemented with ideal current sources. (b) Folded Cascode amplifier, implemented with real current sources.

It is known that the gain of a common-source amplifier is given from the equation:

$$A_u = -g_m R_o \quad (\text{B.1})$$

In the Folded Cascode topology, the common-source transistor provides the amplification, while the cascode transistor enhances the output resistance. Since the

output of the Folded Cascode amplifier is at the drain of the common-gate transistor, the output resistance will be (It is assumed that both current sources are ideal):

$$R_o = g_{m2} r_{ds2} r_{ds1} \quad (\text{B.2})$$

Therefore, the gain of the Folded Cascode amplifier for ideal current sources is:

$$A_u = -g_{m1} g_{m2} r_{ds2} r_{ds1} \quad (\text{B.3})$$

In the circuit that was designed, a PMOS transistor is providing the bias current I_{bias1} , while I_{bias2} is provided by a cascode current source implemented with NMOS transistors. Furthermore, the resistance $I_{bias1}(r_{L1})$ is the internal resistance of transistor M3, and the resistance $I_{bias2}(r_{L2})$ is the resistance of the cascode arrangement (transistors M4 and M5). In the small-signal model, (r_{L1} is in parallel with the resistance of the input transistor, while r_{L2} is in parallel with the output resistance of the Folded Cascode amplifier. Therefore:

$$R_o = [g_{m2} r_{ds2} (r_{ds1} || r_{L1})] || [g_{m3} r_{ds3} r_{ds4}] \quad (\text{B.4})$$

$$A_u = -g_{m1} [g_{m2} r_{ds2} (r_{ds1} || r_{L1})] || [g_{m3} r_{ds3} r_{ds4}] \quad (\text{B.5})$$

Appendix C

Current Mirrors

The following figure depicts the topology of a current mirror implemented with PMOS and NMOS elements, as both types are used in the thesis's preamplifier circuit. The transistor M1 is biased by the current source I_{REF} in both implementations.

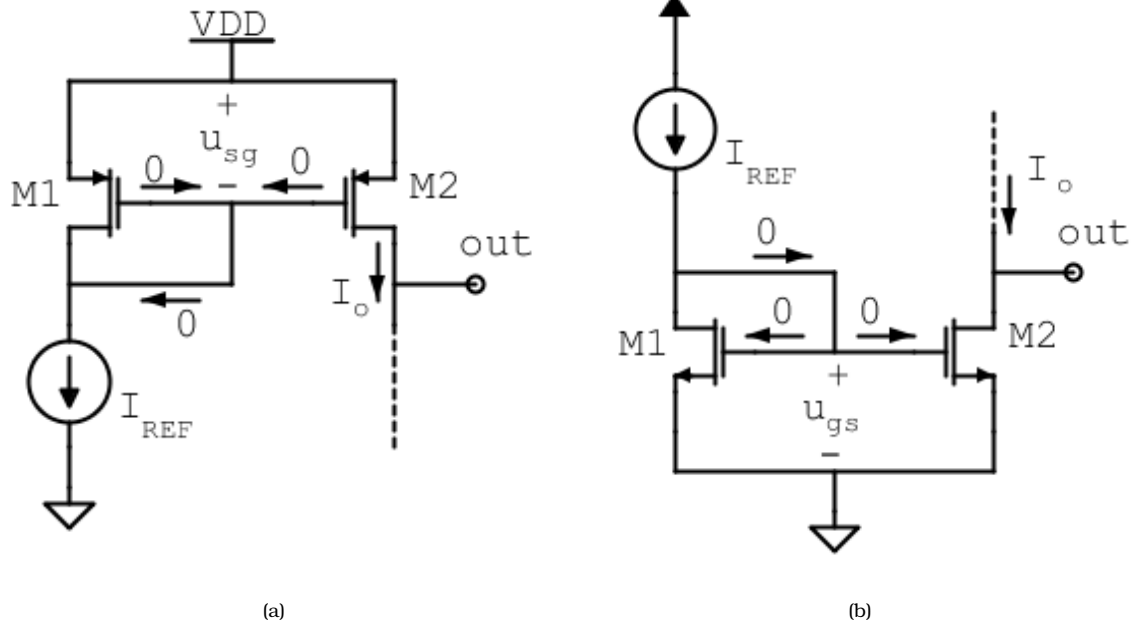


Figure 1: (a) Current mirror implemented with PMOS transistors. (b) Current mirror implemented with NMOS transistors.

In the current mirror with PMOS devices, as well as in that with NMOS devices, the inequalities $u_{SD} \geq |u_{GS} - V_{tp}|$ and $u_{DS} \geq u_{GS} - V_{tn}$, respectively, are satisfied. Therefore, they operate in the saturation region (relation 4.2).

As long as no DC current flows to the gates of the transistors, the source current will be equal to the current flowing through transistor M1. In most cases, reference current (I_{REF}) is implemented by applying a voltage to one end of a resistor (R), while the other end is connected to transistor M1. In this way, by varying the resistance (R), the current flowing through the transistor is controlled.

Lastly, since transistor M2 has the same gate-source voltage as M1 (their gates are connected) and operates in saturation, it will also have a current equal to the expressions of equations (4.2). However, the two transistors (M1 and M2) may have different dimensions. Therefore, by dividing the currents flowing through the transistors, the ratio of the output current to the reference current is obtained.

$$\frac{I_o}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1} \quad (C.1)$$

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